

# THE LEARNING OPERATING SYSTEM

An Enterprise Operating Framework for  
AI-Assisted Learning Systems



Joshua Osondu

# THE LEARNING OPERATING SYSTEM

*Why AI Needs Governance Before It Needs Speed*

**Introducing IDOS**

*(pronounced “eye-doss”)*

*An Enterprise Operating Framework for AI-Assisted Learning Systems*

By

**Joshua Osondu**

Framework Version: IDOS Version 1.0

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AI-Assisted Learning Systems

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Framework composites, case study framing: The applied case studies in Chapters 10 and 11 are composites synthesized from common, well-documented organizational patterns and do not describe any single real organization, individual, or event.

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## AI Use and Responsibility Statement

In keeping with the governance principles this book advocates, I want to be transparent about how AI was used in its production.

Generative AI tools supported this book's development in the following ways: research support, drafting, and editorial refinement. All instructional design decisions, the IDOS framework's architecture, the five operational engines, the twenty-role agent taxonomy, the case studies, and the arguments made throughout this book are my own, developed through my doctoral research, professional practice, and direct experience implementing AI-assisted instructional systems. AI-generated drafts of specific passages were reviewed, verified, and substantially revised by me before inclusion, consistent with the Human Design Engine's own principle that human professionals, not AI, determine intent, structure, and final judgment (Section 1.1).

Where this book cites AI capabilities, risks, or governance practices, those claims are supported by the literature referenced throughout and reflect my own analysis and synthesis of that literature, not AI-generated claims presented as fact.

This statement is offered in the same spirit as the Transparency and Documentation Architecture described in Chapter 2: readers should be able to trust not only the framework a book proposes, but how the book itself came to exist.

*Dedicated to educators, instructional designers, and researchers  
around the world who continue to advance learning through  
curiosity, collaboration, innovation, and responsible leadership.*

## Preface

Artificial intelligence has accelerated many aspects of instructional work. Today, organizations can generate learning content, assessments, multimedia resources, and instructional analyses in minutes rather than weeks (Bolick & Da Silva, 2024; McNeill et al., 2025). Yet while AI has made instructional production faster, it has also exposed a challenge that technology alone cannot solve. Organizations increasingly need a consistent way to govern AI-assisted instructional work, coordinate human and AI responsibilities, maintain instructional quality, preserve organizational knowledge, and improve continuously across projects. For many institutions, these organizational capabilities have not evolved as quickly as the technologies themselves.

Through extensive reading, research, professional practice, and observation across instructional design, faculty development, AI strategy, educational technology, and organizational learning, I noticed a recurring pattern. Organizations were experimenting successfully with AI, but they often did so through isolated tools, individual prompting practices, or disconnected workflows. Teams adopted different AI applications, followed different review processes, documented decisions inconsistently, and rarely captured lessons learned in ways that strengthened future work. The challenge was rarely whether AI could generate instructional materials. The more fundamental challenge was how organizations could integrate AI into instructional work systematically, responsibly, and sustainably.

Existing instructional design methodologies, including ADDIE, the Dick and Carey Systematic Design of Instruction, Backward Design, Successive Approximation Model (SAM), Agile instructional design, and other established approaches, continue to provide valuable guidance for analyzing instructional needs, designing learning

experiences, developing instructional resources, and evaluating learning outcomes (Allen & Sites, 2012; Dick et al., 2014; Molenda, 2003; Wiggins & McTighe, 2005). Their value remains unquestioned. However, they were developed before AI became an integral part of instructional practice, and none was designed to coordinate enterprise AI governance, Human–AI collaboration, AI-assisted instructional production, quality intelligence, organizational knowledge management, and continuous organizational learning within a single operational architecture.

This book introduces the Human–AI Instructional Design Operating System (IDOS) in response to that challenge. IDOS is not another instructional design methodology, nor is it intended to replace established design models. Instead, it provides an enterprise operating framework that enables organizations to implement those methodologies consistently within AI-enabled environments. By coordinating instructional design, AI governance, instructional production, quality validation, and organizational learning through five integrated operational engines, IDOS seeks to provide the organizational architecture needed for responsible, scalable, and sustainable Human–AI instructional work.

One principle guided the development of this framework from the beginning: technology should serve organizational capability, not define it. AI platforms will continue to evolve, and organizations should not need to redesign their instructional governance each time a new model, application, or prompting technique appears. For that reason, IDOS is intentionally platform independent. The framework is designed around enduring organizational capabilities rather than specific technologies, allowing institutions to adopt new AI tools while preserving consistent governance, operational practices, and instructional quality.

Throughout the book, AI is treated as a capability that extends human expertise rather than replacing it. Human professionals remain responsible for instructional intent, governance decisions, ethical judgment, quality standards, and organizational accountability. AI contributes where it offers the greatest value: accelerating analysis, supporting production, identifying patterns, assisting quality review, and strengthening continuous improvement within clearly defined human authority.

This book is written for instructional designers, faculty developers, learning leaders, corporate learning professionals, AI governance practitioners, researchers, and organizational decision-makers seeking to integrate AI into instructional work responsibly and systematically. I offer this framework not as a finished destination but as a foundation for continued discussion, refinement, empirical evaluation, and collaborative development as both instructional design and AI continue to evolve.

Joshua Osondu  
Athens, Ohio, USA, 2026

## Acknowledgments

The Human–AI IDOS builds on decades of scholarship in instructional design, learning sciences, organizational learning, systems thinking, and responsible AI. While this book presents a new enterprise framework, it stands on the contributions of the researchers and practitioners whose work has advanced these fields, many of whom are cited throughout.

I am grateful to the faculty mentors, colleagues, and academic advisors whose scholarship and feedback challenged me to think more carefully about instructional design, learning technologies, and organizational change, and to the faculty, staff, and colleagues at Ohio University who provided the interdisciplinary environment in which many of these ideas were first tested.

I also want to acknowledge the many instructional designers, educators, administrators, and technology specialists whose conversations and challenges shaped my understanding of how organizations are actually navigating AI adoption, not how the literature assumes they are. Their questions, more than any single source, pushed me toward a more systematic approach.

Finally, my deepest gratitude goes to my family and friends for their patience and encouragement throughout this project. Any errors, omissions, or interpretive choices that remain are entirely my own.

# Introduction

## *Why Organizations Need a Human–AI Instructional Design Operating System*

Artificial intelligence (AI) is becoming a routine component of how organizations design, develop, deliver, and continuously improve learning, assisting with content development, assessment design, data analysis, accessibility, and quality review across higher education, corporate learning, government, and nonprofit settings (Chng, 2023; Pelletier et al., 2024). These advances offer significant opportunities to improve productivity, scalability, and access to learning (Bolick & Da Silva, 2024; Holmes & Tuomi, 2022). They also introduce new questions about governance, accountability, instructional quality, and the role of human expertise that many organizations continue to address on a project-by-project basis rather than through coordinated enterprise practices.

This book argues that successful AI integration is fundamentally an organizational challenge rather than a technology challenge. Organizations do not simply need new AI applications; they need an enterprise operating framework that coordinates governance, operations, Human–AI collaboration, quality management, and organizational learning. Without one, AI initiatives tend to stay fragmented, inconsistent across teams, difficult to govern, and hard to sustain.

The Human–AI IDOS was developed to address three recurring gaps in organizational AI-assisted instructional work. First, the instructional gap: AI can generate instructional materials but cannot independently determine instructional intent, establish learning outcomes, or judge whether a learning experience achieves its intended goals. These responsibilities require human expertise from the outset (Bolick & Da Silva, 2024; Dick et al., 2014). Second, the operational gap: most

organizations lack a standardized way to integrate AI into instructional work, so teams use different tools, workflows, and review practices, and consistency suffers (Bolick & Da Silva, 2024; McNeill et al., 2025). Third, the organizational capability gap: even successful instructional projects rarely build lasting organizational capability without a structured way to capture what was learned and reuse it (Argyris & Schön, 1978; DiBella et al., 1996).

Rather than introducing another instructional design methodology, IDOS provides the enterprise architecture that enables AI-assisted instructional work to be designed, governed, produced, validated, and continuously improved as a coordinated organizational system. IDOS addresses these gaps through five integrated operational engines, Human Design, AI Governance, AI Instructional Production, Quality Intelligence, and Knowledge Intelligence, introduced in Chapter 2, and through four enterprise management documents that support implementation, introduced in Part II. Together, these operational engines establish a governed workflow that begins with human instructional intent, continues through AI-assisted production and independent quality validation, and concludes with organizational knowledge capture to support continuous improvement.

Throughout the framework, AI is treated as a capability that extends human performance rather than replaces it: human professionals remain responsible for instructional goals, professional judgment, and instructional quality, while AI supports analysis, production, and continuous improvement within governance boundaries they establish. IDOS does not replace instructional design methodologies such as ADDIE, Backward Design, and the Dick and Carey systematic design of instruction; it provides the enterprise architecture within which they can operate consistently and responsibly in AI-enabled environments (Dick et al., 2014; Molenda, 2003; Wiggins & McTighe, 2005). Although the applied examples in this book draw on specific

organizational contexts, the underlying principles of governance, Human–AI collaboration, operational coordination, instructional quality, and organizational learning are intended to generalize across organizations that design, deliver, and continuously improve learning.

This book is organized into four parts. Part I establishes the conceptual foundations of IDOS, presents the five operational engines that form its execution architecture, and explains how those engines function together as one integrated system, situating the framework relative to existing instructional-design scholarship along the way. Part II provides practical guidance for implementation, governance, operation, and continuous improvement, organized around four reusable enterprise management documents. Part III extends the framework to two enterprise instructional practices: designing new instructional systems and modernizing existing instructional resources. Part IV demonstrates the framework through two applied case studies: a corporate compliance training redesign and a higher education AI practitioner certification program. Both cases are composites synthesized from common organizational patterns, allowing IDOS to be evaluated in two different sectors without relying on a single organizational implementation. Reusable templates, a glossary, and a candid discussion of the framework's current limitations follow in the back matter.

A note on the name: the echo of “OS” in IDOS is deliberate, not incidental. Early computer operating systems succeeded because they provided a common layer through which software applications could reliably access memory, storage, process scheduling, and processing resources without redefining those capabilities for every application (Hansen, 1973). This book argues that instructional design has not yet converged on a widely adopted enterprise operating architecture for AI that integrates governance, quality assurance, organizational learning, standardized operational workflows, and lifecycle

management within a single enterprise operating architecture. The central claim of this book is that instructional design now requires the equivalent of that foundational layer, and the name is intended to communicate that idea directly rather than merely allude to it.

A note on AI: AI technologies will continue to evolve rapidly, and some tools named or implied in this book may be superseded by the time it is read. For that reason, IDOS is intentionally platform independent. The framework is built around enduring organizational capabilities, including governance, Human–AI collaboration, instructional quality, operational coordination, and organizational learning, rather than dependence on any particular AI technology. As AI capabilities evolve, organizations should be able to adopt new tools without redesigning their underlying instructional operating model. The technologies will change; the organizational principles presented in this book are intended to endure.

## About the Author

Joshua Osondu is an instructional designer, AI education researcher, and learning systems strategist whose work focuses on the intersection of instructional design, AI, organizational learning, implementation-science, and educational innovation. His doctoral research centers on the Generic AI Competency (GAIC) Scale, an evidence-based instrument for measuring applied AI competency. His research and professional practice examine how organizations can integrate AI into learning and instructional systems in ways that are systematic, governed, human-centered, and sustainable.

He is the founder of the Nexus Institute for AI Learning, Inc., a nonprofit organization advancing AI competency frameworks, competency-based learning systems, and implementation-science research supporting responsible AI adoption in education and the workforce.

He is a Ph.D. Candidate in Innovative Learning Design and Technology at Ohio University, where he also earned an M.Ed. in Educational Research and Evaluation and an M.A. in International Studies (African Studies). He received a B.A. in History and Political Science from Kwame Nkrumah University of Science and Technology (KNUST), Ghana.

Joshua has designed and evaluated online and blended learning experiences, developed competency-based learning systems, contributed to institutional AI literacy and governance initiatives, and led instructional innovation projects in higher education. His professional work spans instructional design, learning experience design, instructional technology, AI governance and policy, and sustainability across the curriculum.

The Human–AI Instructional Design Operating System (IDOS) presented in this book reflects Joshua's interdisciplinary background in instructional design, organizational research, AI governance, and systems thinking. Rather than viewing AI as simply another instructional technology, his work explores how organizations can establish enterprise capabilities that integrate governance, quality

assurance, organizational learning, and human–AI collaboration into a coherent operating framework for instructional design and learning systems – the framework he presents in this book to help organizations integrate AI into instructional work while preserving human expertise and continuous organizational learning.

# How to Use This Book

The Human–AI IDOS is designed to serve a broad audience – executives, instructional designers, faculty, AI governance leads, researchers, and organizations planning implementation. Every reader benefits from the Introduction and Part I; beyond that, the table below suggests a path suited to your role. Chapters not visited on a shorter path remain useful as reference and can be read at any time.

| If you are...                   | Recommended path                                                                                                                                                                     |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Executive leader / CLO          | Introduction → Part I → Ch. 3 (Integration) → Ch. 5 (Governance) → Appendix B (Quick Guide)                                                                                          |
| Instructional designer          | Read sequentially, Parts I–IV — Part IV (Case Studies) shows the framework applied end to end, in two sectors                                                                        |
| Faculty / instructor            | Introduction → Part I → Ch. 6 (Operations) → Ch. 11 (AI Practitioner Certification Case) → Appendix B                                                                                |
| AI governance / compliance lead | Introduction → Ch. 2 (AI Governance Engine) → Ch. 5 (Governance) → Ch. 10 or Ch. 11 (both show governance catching a real category of error) → Appendix A (Governance Plan template) |
| Researcher / graduate student   | Full sequential read, Parts I–IV, plus Limitations and Future Research and full References                                                                                           |
| Organization implementing IDOS  | Introduction → Part I in full → Part II in full → Appendix A templates → Part IV as a worked reference                                                                               |

Regardless of path, organizations planning to adopt IDOS should treat this book as both a reference and a working implementation guide: read Part I for the conceptual and architectural foundation, use Part II to plan governance and operations, apply the templates in Appendix A during implementation, and consult Part IV's worked case when questions arise about how a specific artifact should look in practice.

A note on terminology: this book uses several related but distinct terms, framework, operating system, architecture, engine, operational

object, governance, and workflow, precisely rather than interchangeably, and a few of them (governance, workflow) deliberately carry both a general meaning and a specific, capitalized one. Appendix C's glossary defines each term individually; a short terminology map at the start of that appendix shows how they relate to one another. Reading it before your first pass through Part I is worth the five minutes, particularly if any of these words already carry a specific meaning inside your own organization.

# Table of Contents

AI Use and Responsibility Statement.....iii

Preface ..... v

Acknowledgments.....viii

Introduction .....ix

About the Author.....xiii

How to Use This Book .....xv

Table of Contents.....xvii

Foundations and Architecture..... 1

Chapter 1 .....2

    1.1 What Is IDOS?.....2

    1.2 Why IDOS Was Developed .....3

*The instructional gap* .....4

*The operational gap*.....4

*The organizational gap*.....4

    1.3 Scope of IDOS.....5

    1.4 Core Capabilities of IDOS .....6

    1.5 Enterprise Architecture of IDOS.....7

    1.6 Foundational Design Principles.....8

    1.7 Positioning IDOS Within Instructional Design .....9

Chapter 2.....12

    2.1 One Architecture, Five Engines .....12

    2.2 Engine-by-Engine Comparison.....14

    2.3 Human Design Engine – The Ten Human Design Domains.....16

    2.4 AI Governance Engine – The Ten Governance Domains .....18

    2.5 AI Instructional Production Engine – The Nine Instructional Production Domains .....20

|                                                                    |    |
|--------------------------------------------------------------------|----|
| 2.6 Quality Intelligence Engine – The Ten Quality Domains .....    | 21 |
| 2.7 Knowledge Intelligence Engine – The Ten Knowledge Domains..... | 23 |
| 2.8 How the Five Engines Depend on One Another .....               | 25 |
| 2.9 The Cross-Engine AI Agent Taxonomy .....                       | 27 |
| 2.9.1 Enterprise Prompt Architecture.....                          | 32 |
| 2.10 A Note on Recurring Numbers .....                             | 35 |
| 2.11 Chapter Summary .....                                         | 36 |
| Chapter 3.....                                                     | 38 |
| 3.1 The Operational Objects as a Single Source of Truth.....       | 38 |
| 3.2 Human–AI Collaboration Across the System .....                 | 39 |
| 3.3 Governance Is Distributed, Not Centralized in One Engine ..... | 40 |
| 3.4 Why IDOS Is Different — Revisited.....                         | 40 |
| 3.5 Chapter Summary.....                                           | 41 |
| Implementation & Governance .....                                  | 42 |
| Chapter 4.....                                                     | 43 |
| 4.1 Why Readiness Assessment Comes First.....                      | 43 |
| 4.2 The Five Readiness Pillars.....                                | 43 |
| 4.3 The Readiness Maturity Model.....                              | 46 |
| 4.4 Prioritizing Capability Gaps.....                              | 47 |
| 4.5 The Four-Phase Implementation Roadmap .....                    | 48 |
| 4.6 Common Implementation Risks.....                               | 49 |
| 4.7 Chapter Summary.....                                           | 50 |
| Chapter 5.....                                                     | 51 |
| 5.1 From Project Governance to Organizational Governance .....     | 51 |
| 5.2 The Five Governance Domains.....                               | 51 |
| 5.3 Decision Authority .....                                       | 52 |

|                                                                                             |           |
|---------------------------------------------------------------------------------------------|-----------|
| 5.4 Governance Documentation .....                                                          | 54        |
| 5.5 Enterprise AI Policy Framework .....                                                    | 54        |
| 5.6 Monitoring Governance Effectiveness .....                                               | 56        |
| 5.7 Chapter Summary.....                                                                    | 56        |
| <b>Chapter 6.....</b>                                                                       | <b>58</b> |
| 6.1 The Six-Stage Enterprise Workflow .....                                                 | 58        |
| 6.2 Designing Workflows That Apply These Principles<br>Rather Than Copying a Template ..... | 59        |
| 6.3 Coordinating the Five Engines in Daily Operation.....                                   | 60        |
| 6.4 Monitoring Operational Execution.....                                                   | 61        |
| 6.5 Continuous Operational Adaptation.....                                                  | 61        |
| 6.6 Chapter Summary.....                                                                    | 61        |
| <b>Chapter 7.....</b>                                                                       | <b>63</b> |
| 7.1 Why Measure the System, Not Just the Projects .....                                     | 63        |
| 7.2 The Six Performance Domains .....                                                       | 63        |
| 7.3 Evaluating the Five Operational Engines.....                                            | 64        |
| 7.4 Indicators, Dashboards, and Decision Support.....                                       | 65        |
| 7.5 From Measurement to Organizational Learning .....                                       | 65        |
| 7.6 Enterprise Performance Reviews.....                                                     | 66        |
| 7.7 Chapter Summary.....                                                                    | 66        |
| <b>Designing and Modernizing Learning Systems.....</b>                                      | <b>67</b> |
| <b>Chapter 8.....</b>                                                                       | <b>68</b> |
| 8.1 Six Design Objects, One Workflow .....                                                  | 68        |
| 8.2 Course and Module Design .....                                                          | 70        |
| 8.3 Lesson Design — The IDOS Lesson Flow .....                                              | 71        |
| 8.4 Learning Pathway Design — The IDOS Learner Journey<br>.....                             | 72        |
| 8.5 Competency-Based Program Design – The IDOS<br>Competency Mastery Framework.....         | 73        |

|                                                                                          |     |
|------------------------------------------------------------------------------------------|-----|
| 8.6 Microlearning and Performance Support – The IDOS Performance Support Framework ..... | 74  |
| 8.7 One Pattern, Four Names .....                                                        | 75  |
| 8.8 Chapter Summary .....                                                                | 76  |
| Chapter 9 .....                                                                          | 77  |
| 9.1 Guiding Principles of Instructional Modernization .....                              | 77  |
| 9.2 The Instructional Modernization Maturity Model .....                                 | 78  |
| 9.3 Legacy Asset Classification .....                                                    | 79  |
| 9.4 The Instructional Modernization Lifecycle .....                                      | 80  |
| 9.5 Instructional Modernization Risk .....                                               | 82  |
| 9.6 Instructional Modernization Quality Assurance .....                                  | 83  |
| 9.7 AI Governance for Modernization .....                                                | 83  |
| 9.8 Chapter Summary .....                                                                | 83  |
| Applied Case Studies .....                                                               | 85  |
| Chapter 10 .....                                                                         | 86  |
| 10.1 Case Overview .....                                                                 | 86  |
| 10.2 Stage 1: Human Design Engine .....                                                  | 87  |
| 10.3 Stage 2: AI Governance Engine .....                                                 | 89  |
| 10.4 Stage 3: AI Instructional Production Engine – A Governed Draft in Practice .....    | 90  |
| 10.5 Stage 4: Quality Intelligence Engine .....                                          | 92  |
| 10.6 Stage 5: Knowledge Intelligence Engine .....                                        | 94  |
| 10.7 A Second Design Object: The Follow-Up Job Aid .....                                 | 95  |
| 10.8 Before/After: Governed vs. Ad Hoc AI Use .....                                      | 96  |
| 10.9 What This Case Does, and Does Not, Show .....                                       | 97  |
| 10.10 Chapter Summary .....                                                              | 98  |
| Chapter 11 .....                                                                         | 99  |
| 11.1 Case Overview .....                                                                 | 99  |
| 11.2 Stage 1: Human Design Engine .....                                                  | 101 |
| 11.3 Stage 2: AI Governance Engine .....                                                 | 102 |

|                                                                                          |     |
|------------------------------------------------------------------------------------------|-----|
| 11.4 Stage 3: AI Instructional Production Engine – A<br>Governed Draft in Practice ..... | 104 |
| 11.5 Stage 4: Quality Intelligence Engine .....                                          | 106 |
| 11.6 Stage 5: Knowledge Intelligence Engine .....                                        | 108 |
| 11.7 Before/After: Governed vs. Ad Hoc AI Use.....                                       | 109 |
| 11.8 What This Case Does, and Does Not, Show .....                                       | 110 |
| 11.9 Chapter Summary .....                                                               | 111 |
| Appendices, References, and Index.....                                                   | 112 |
| Appendix A.....                                                                          | 113 |
| A.1 Enterprise Readiness Assessment (ERA) .....                                          | 113 |
| A.2 Enterprise Governance Plan (EGP) .....                                               | 114 |
| A.3 Operational Execution Framework (OEF).....                                           | 115 |
| A.4 Enterprise Performance Management Plan (EPMP)....                                    | 116 |
| Appendix B.....                                                                          | 117 |
| B.1 The Seven-Step Sequence .....                                                        | 117 |
| B.2 Before You Begin .....                                                               | 117 |
| B.3 Implementation Checklist.....                                                        | 118 |
| Appendix C.....                                                                          | 119 |
| How These Terms Relate .....                                                             | 119 |
| Limitations and Future Research.....                                                     | 125 |
| What This Book Has Not Shown.....                                                        | 125 |
| Scope Limitations.....                                                                   | 127 |
| Future Research Directions.....                                                          | 128 |
| A Closing Note.....                                                                      | 128 |
| References.....                                                                          | 130 |
| Index.....                                                                               | 137 |

HUMAN–AI IDOS

PART I

# Foundations and Architecture

## Chapter 1

# Introducing the Human–AI Instructional Design Operating System (IDOS)

*This chapter introduces IDOS and establishes the conceptual foundation for the rest of the book: why it was developed, what it covers, how its five operational engines relate to one another, the principles that guide it, and how it is positioned relative to existing instructional design scholarship. Later parts examine each operational engine, the integrated architecture, and implementation in detail.*

## 1.1 What Is IDOS?

The Human–AI IDOS is an enterprise framework for designing, governing, producing, implementing, evaluating, and continuously improving instructional systems through coordinated Human–AI collaboration. Rather than treating AI as an independent instructional designer, IDOS establishes a governed operating environment in which human professionals determine instructional goals, strategies, and quality expectations, and AI accelerates production, automates selected tasks, and assists with implementation within clearly defined governance boundaries.

The framework extends beyond the development of individual learning experiences to coordinate the complete instructional lifecycle, from identifying an instructional need to improving organizational capability over time, through five coordinated operational engines, summarized in Table 1.1.

| Operational engine        | Primary responsibility                                                          |
|---------------------------|---------------------------------------------------------------------------------|
| Human Design Engine (HDE) | Instructional intent, learning outcomes, strategy, and assessment requirements. |

|                                           |                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------|
| AI Governance Engine (AGE)                | Policies, operating rules, and oversight for Human–AI collaboration.           |
| AI Instructional Production Engine (AIPE) | AI-supported production of instructional resources under human supervision.    |
| Quality Intelligence Engine (QIE)         | Instructional quality, accessibility, and implementation-readiness validation. |
| Knowledge Intelligence Engine (KIE)       | Organizational learning, reusable knowledge, and continuous improvement.       |

*Table 1.1. The five operational engines of IDOS.*

Each engine has a distinct responsibility, but none operates independently: decisions made in one guide the work performed in the next, forming a coordinated operating environment that maintains instructional quality, governance, and organizational consistency across the lifecycle. Chapter 2 examines all five engines together as a single comparative model.

IDOS is intentionally platform-independent. It is not tied to a specific AI application, learning management system, or vendor – organizations can adopt new technologies without changing the framework, because instructional decisions are established before AI-assisted production begins. At its core, IDOS rests on one principle: human expertise directs the instructional system, and AI extends organizational capability.

## 1.2 Why IDOS Was Developed

Generative AI can now produce lesson plans, assessments, videos, and facilitator guides in minutes. That capability alone does not constitute an instructional system. Many organizations have adopted AI as a content-generation tool without establishing consistent governance, quality assurance, or organizational-learning processes around it, and as adoption spreads across teams and departments, the resulting inconsistency, duplicated effort, and limited oversight compound. The

open question is not whether AI can generate instructional content; it is whether organizations can manage AI-assisted instructional work consistently, reliably, and accountably. IDOS was developed to close three related gaps.

### *The instructional gap*

AI can generate materials, but it cannot independently determine instructional intent, analyze learner needs, or judge whether a learning experience achieves its intended outcome. IDOS addresses this by requiring that instructional decisions be established by human professionals before AI-assisted production begins — consistent with longstanding instructional-design scholarship holding that systematic analysis and design must precede development (Dick et al., 2014; Gagné et al., 2005).

### *The operational gap*

Most organizations lack a standardized approach for integrating AI into instructional work: teams use different tools, workflows, and review practices, which makes consistency across projects difficult to sustain (Bolick & Da Silva, 2024; McNeill et al., 2025). IDOS addresses this by providing a common operational architecture for Human–AI collaboration, governance, and production documentation (detailed in Parts II and IV).

### *The organizational gap*

Successful instructional projects do not automatically build organizational capability. Without a structured process for capturing knowledge and evaluating performance, organizations tend to repeat work and lose valuable experience (Argyris & Schön, 1978; DiBella et al., 1996). IDOS addresses this by integrating quality intelligence and knowledge management into the operating system itself, rather than treating organizational learning as an informal, optional afterthought.

Together, these gaps reframe the central question. Rather than “how can AI generate instructional content,” IDOS asks how organizations can design, govern, produce, implement, evaluate, and continuously improve AI-assisted instructional systems, a shift from content generation to enterprise instructional capability that runs through the rest of this book.

### 1.3 Scope of IDOS

IDOS organizes its scope into five complementary domains: instructional design (analysis, design, development, implementation, and evaluation); Human–AI collaboration and governance (decision authority, accountability, and oversight); enterprise instructional operations (coordinated workflows and reusable operational objects); technology integration (a platform-independent approach to AI services and learning technologies); and organizational capability development (governance, quality assurance, and knowledge management that outlast any single project).

Table 1.2 maps this scope across the instructional lifecycle, from initial analysis through continuous improvement, illustrating that IDOS coordinates a connected system rather than a set of isolated activities.

| Lifecycle stage | Primary enterprise focus                                                           |
|-----------------|------------------------------------------------------------------------------------|
| Analysis        | Identify instructional problems, learner needs, and organizational requirements.   |
| Design          | Define learning outcomes, strategies, assessments, and accessibility requirements. |
| Production      | Develop instructional resources through governed AI-assisted workflows.            |
| Implementation  | Deliver instruction, support learners, and manage operational readiness.           |
| Evaluation      | Measure effectiveness, quality, governance compliance, and performance.            |

|                        |                                                                |
|------------------------|----------------------------------------------------------------|
| Continuous improvement | Capture knowledge and strengthen future instructional systems. |
|------------------------|----------------------------------------------------------------|

*Table 1.2. IDOS across the instructional lifecycle.*

IDOS provides the operating environment for these activities; it does not replace subject-matter expertise, institutional policy, legal and regulatory compliance, professional standards, or organizational governance. Those responsibilities remain with the organization – IDOS exists to help carry them out consistently across instructional initiatives.

## 1.4 Core Capabilities of IDOS

Each operational engine delivers one core enterprise capability. The output of one becomes the input to the next, creating a continuous workflow rather than five disconnected services (Table 1.3).

| Capability                           | Delivered by                       |
|--------------------------------------|------------------------------------|
| Human-centered instructional design  | Human Design Engine                |
| Human-AI governance                  | AI Governance Engine               |
| AI-assisted instructional production | AI Instructional Production Engine |
| Quality intelligence                 | Quality Intelligence Engine        |
| Knowledge intelligence               | Knowledge Intelligence Engine      |

*Table 1.3. Core capabilities and their primary operational engine.*

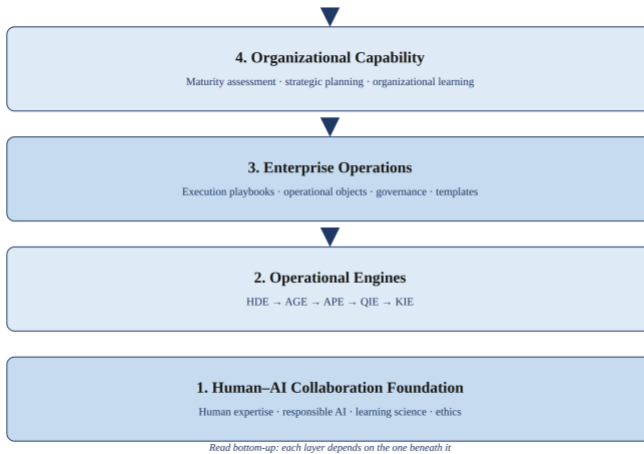
The flow is iterative, not strictly linear: findings from quality evaluation or organizational learning frequently require revisiting instructional design, governance rules, or production workflows. This is what allows IDOS to improve both instructional products and organizational capability over time, rather than treating each project as a closed loop.

## 1.5 Enterprise Architecture of IDOS

IDOS is organized as a four-layer architecture. Each layer supports the layers above it and relies on the layers below it, which allows organizations to manage instructional systems consistently across projects, departments, and technology platforms (Table 1.4).

| Layer                                | Contains                                                                                                                             |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 4. Organizational capability         | Maturity assessment, strategic planning, organizational learning, and continuous improvement.                                        |
| 3. Enterprise operations             | Enterprise management documents, operational specifications, governance documentation, workflows, templates, and enterprise outputs. |
| 2. Operational engines               | HDE → AGE → AIPE → QIE → KIE (Chapter 2).                                                                                            |
| 1. Human-AI collaboration foundation | Human and organizational expertise, learning sciences, responsible AI, ethics, and organizational values.                            |

*Table 1.4. The four-layer enterprise architecture of IDOS, from foundation to organizational capability.*



*Figure 1.1. The same four layers, read bottom-up: each layer depends on the one beneath it, from the Human-AI collaboration foundation through to organizational capability.*

The foundation layer establishes that instructional professionals, not AI, define instructional intent, learner needs, and success criteria, informed by human organizational expertise, learning science and responsible-AI principles (Shneiderman, 2022). The operational engine layer executes instructional work through the five engines introduced in Section 1.1. The enterprise operations layer supplies the templates, playbooks, and governance documentation that let different teams apply the same operating principles consistently (see Part II and Appendix A). The organizational capability layer looks beyond any single project to organizational learning, maturity, and continuous improvement, ensuring that instructional initiatives build lasting capability rather than ending when a deliverable ships.

## 1.6 Foundational Design Principles

Ten principles guide how IDOS is designed, implemented, governed, and improved. Unlike operational procedures, which vary by organization, these principles are intended to remain constant regardless of instructional context or technology platform.

- Human instructional authority – human organizational expertise directs the instructional system; AI supports but does not replace professional judgment (Shneiderman, 2022).
- Governed Human–AI collaboration – operationalizes human instructional authority by defining how AI supports instructional work while preserving human responsibility and accountability (McNeill et al., 2025; National Institute of Standards and Technology [NIST], 2023; Yilmaz, 2025).
- Learner-centered design – instructional decisions begin with learner needs, not available technology (Dick et al., 2014).
- Instructional alignment – outcomes, strategies, activities, assessments, and resources remain aligned throughout the lifecycle (Biggs, 1996).

- Enterprise governance – consistent policies, standards, and oversight support responsible, repeatable operations (Martins, 2026).
- Evidence-informed decision-making – instructional decisions rest on research, learner data, and evaluation findings, not assumption (Dick et al., 2014; Gagné et al., 2005).
- Organizational learning – every project should strengthen the ones that follow it, not just itself (Argyris & Schön, 1978; DiBella et al., 1996).
- Platform independence – the framework's operating principles do not depend on any specific AI application or vendor (Alghazzawi et al., 2021).
- Continuous improvement – quality evaluation and organizational learning inform ongoing refinement (Oliver, 2009).
- Enterprise scalability – the same operating principles apply whether IDOS governs a single module or an enterprise-wide program (Abbott & Fisher, 2009).

These principles are not independent of one another: human authority establishes the foundation for decision-making, governance defines how Human–AI collaboration occurs within that authority, learner-centered design and instructional alignment protect quality, and the remaining principles, evidence, learning, platform independence, continuous improvement, and scalability, keep the system adaptable as organizations and technologies change.

## 1.7 Positioning IDOS Within Instructional Design

IDOS builds on established instructional-design theory rather than replacing it. Models such as ADDIE (Molenda, 2003), Backward Design (Wiggins & McTighe, 2005), the Successive Approximation Model (Allen & Sites, 2012), Agile instructional design (adapted from

Beck et al., 2001), and rapid-prototyping approaches (Tripp & Bichelmeyer, 1990) remain essential for analyzing learning needs, designing instruction, and evaluating learner performance. Most, however, were developed before AI became integral to instructional work, and none was designed to coordinate enterprise governance, Human–AI collaboration, organizational knowledge management, or continuous improvement across an entire organization.

As organizations increasingly integrate AI into instructional work, they need more than an instructional-design methodology, they need a structured operating environment that coordinates Human–AI collaboration, standardizes enterprise workflows, supports quality assurance, preserves organizational knowledge, and enables continuous improvement. IDOS is deliberately methodology-neutral: organizations may continue using ADDIE, Backward Design, SAM, Agile instructional design, or an institution-specific process within the Human Design Engine (Chapter 2). IDOS does not prescribe how instructional design should be performed; it provides the enterprise operating environment within which those methodologies can be implemented consistently and responsibly.

Table 1.5 summarizes the principal differences between conventional instructional design and the enterprise operating environment IDOS provides.

| Conventional instructional design | Human–AI IDOS                                                 |
|-----------------------------------|---------------------------------------------------------------|
| Focuses on designing instruction  | Coordinates the complete instructional operating environment  |
| Human-centered workflows          | Governed Human–AI collaboration                               |
| Limited enterprise governance     | Integrated instructional and AI governance                    |
| Project-specific documentation    | Standardized operational objects and enterprise documentation |

|                                                   |                                                               |
|---------------------------------------------------|---------------------------------------------------------------|
| Evaluation focuses on instructional effectiveness | Continuous quality intelligence and organizational evaluation |
| Knowledge captured inconsistently                 | Systematic organizational knowledge management                |
| Improvement focuses on individual projects        | Continuous organizational capability development              |

*Table 1.5. Conventional instructional design compared with the Human–AI IDOS enterprise operating environment.*

The contribution of IDOS, in short, is not a new instructional-design methodology but an enterprise operating system for Human–AI instructional work, combining instructional design, enterprise governance, AI-assisted production, quality intelligence, knowledge management, and organizational maturity so that organizations can manage instructional systems as long-term strategic capabilities rather than isolated projects. Section 3.4 revisits this positioning once the full architecture has been presented; see the back matter's “Limitations and Future Research” for what remains unproven.

## Chapter 2

### The Five Operational Engines: A Comparative Model

*Chapter 1 introduced the five operational engines individually. This chapter treats them as one system, built on a shared architecture, so that what is common to all five is explained once and what is distinct to each is easy to compare. Organizations implementing IDOS will find this chapter most useful as a reference: return to Section 2.2 for the master comparison table, to Sections 2.3–2.7 for the specific domain checklist of whichever engine is currently in front of you, and to Section 2.9 for the AI agent roles that execute work within each engine.*

#### 2.1 One Architecture, Five Engines

Every operating system needs a kernel, an authoritative source that establishes the rules the rest of the system follows. Within IDOS, the Human Design Engine plays that role for instructional work: before AI generates a lesson plan, assessment, or facilitator guide, the Human Design Engine establishes the instructional decisions that guide everything downstream. This ordering, human instructional reasoning before AI-assisted production, is the single most consequential design choice in the framework, and it is why IDOS begins with design rather than governance or production.

Why five engines, and not four, or six? Because five distinct enterprise questions must be answered before AI-assisted instructional work can be trusted, and each question requires a different kind of authority to answer it. Knowing what each engine is for is the fastest way to know which one a given problem actually belongs to:

| Enterprise question | Answered by |
|---------------------|-------------|
|---------------------|-------------|

|                                                                                |                                           |
|--------------------------------------------------------------------------------|-------------------------------------------|
| What should be designed, and why?                                              | Human Design Engine (HDE)                 |
| Under what rules may AI participate?                                           | AI Governance Engine (AGE)                |
| How are the approved instructional assets actually produced?                   | AI Instructional Production Engine (AIPE) |
| Are those assets ready for implementation?                                     | Quality Intelligence Engine (QIE)         |
| How does the organization get better from every initiative, not just this one? | Knowledge Intelligence Engine (KIE)       |

No engine answers more than one of these questions, and no question is answered by more than one engine. That one-to-one mapping is also why Section 2.8's cross-engine dependencies matter: a change to one answer routinely requires revisiting another, precisely because the five questions are related, not because any one engine has strayed into another's territory.

Each of the five engines shares an identical internal anatomy, summarized in Table 2.1. Understanding this shared anatomy once means that Sections 2.3–2.7 do not need to repeat it, they cover only what is genuinely distinct about each engine's subject matter.

| Shared component            | What it does                                                                                                                                                                                                                   |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Governing philosophy        | A small set of operating commitments specific to that engine's function.                                                                                                                                                       |
| Authoritative specification | The single document of record that engine produces (e.g., the Human Design Specification).                                                                                                                                     |
| Domains                     | The sub-areas of decision-making the specification must address, ten for four of the five engines; nine for AI Instructional Production (Section 2.5), which does not carry a domain duplicating Section 2.9's agent taxonomy. |
| Authority boundary          | What decisions require human sign-off versus what AI may execute unsupervised.                                                                                                                                                 |
| Agent roles                 | Four named AI functional roles per engine, executing work within that engine's authority boundary (Section 2.9).                                                                                                               |

|                              |                                                                                                                                                                              |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Enterprise handoff Artifacts | The standard governance artifacts transferred to the next engine: authoritative specification, supporting package, decision log, traceability matrix, and approved baseline. |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Table 2.1. The shared anatomy common to all five operational engines.

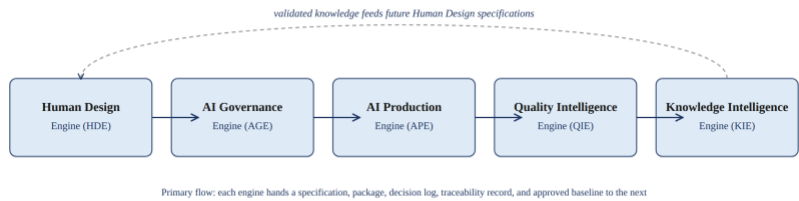


Figure 2.1. The five engines in their primary sequence. The dashed return line is the Knowledge Intelligence Engine feeding validated experience back into the next initiative's Human Design Specification – the mechanism behind the organizational-learning claim in Section 2.7.

2.2 Engine-by-Engine Comparison

Table 2.2 is the single most useful reference in this chapter: it replaces what would otherwise be five long, repetitive chapters with one page that lets a reader compare engines directly.

| Engine                             | Specification                                    | Primary output                         | Human vs. AI authority                                           |
|------------------------------------|--------------------------------------------------|----------------------------------------|------------------------------------------------------------------|
| Human Design (HDE)                 | Human Design Specification (HDS)                 | Approved instructional design baseline | Human: full authority. AI: drafting support only.                |
| AI Governance (AGE)                | AI Governance Specification (AGS)                | Approved governance baseline           | Human: policy and exceptions. AI: compliance monitoring support. |
| AI Instructional Production (AIPE) | AI Instructional Production Specification (AIPS) | Production-ready instructional assets  | AI: generation within rules.                                     |

|                              |                                            |                                      |                                                                                      |
|------------------------------|--------------------------------------------|--------------------------------------|--------------------------------------------------------------------------------------|
|                              |                                            |                                      | Human: review and approval.                                                          |
| Quality Intelligence (QIE)   | Quality Intelligence Specification (QIS)   | Implementation-readiness decision    | Human: independent validation authority. AI: evidence analysis support.              |
| Knowledge Intelligence (KIE) | Knowledge Intelligence Specification (KIS) | Reusable enterprise knowledge assets | Human: approves what becomes organizational knowledge. AI: pattern analysis support. |

*Table 2.2. Comparative summary of the five operational engines.*

Two patterns are worth naming explicitly because they recur throughout the rest of this book. First, authority consistently sits with people, while AI consistently sits inside a support role, drafting, monitoring, analyzing, that a human reviews before it becomes binding (a division consistent with current human-centered AI and AI-governance guidance; see NIST, 2023; Shneiderman, 2022). Second, Every engine hands off the same categories of enterprise handoff artifacts, an authoritative specification, supporting package, decision log, traceability matrix, and approved baseline, even though the specific contents of those artifacts differ from one engine to the next, which is what allows Chapter 3 to trace one continuous chain of evidence from instructional intent through to organizational learning without requiring each subsequent engine to recreate or re-authorize decisions that have already been approved by the preceding engine.

Table 2.2's "Primary output" column names what each engine produces; what it does not show is that each output is the next engine's starting input, not just a milestone marker. Stated as a single chain: the Human Design Specification becomes the AI Governance Specification's starting input, which becomes the boundary the AI

Instructional Production Engine drafts within, whose instructional assets become what the Quality Intelligence Engine validates, whose validation decision becomes what the Knowledge Intelligence Engine captures, whose validated knowledge becomes the next initiative's Human Design Specification. Section 3.1 (Table 3.1) formalizes this dependency chain object by object; the point to take from it here is simpler: nothing in this chain is optional or parallel. An engine that starts before the previous one's output is approved is not moving faster, it is guessing what that output will say.

## 2.3 Human Design Engine – The Ten Human Design Domains

The Human Design Specification is organized into ten domains that, together, define instructional intent before AI participates. This structure reflects long-standing instructional-design practice, needs analysis, learner analysis, outcome specification, and assessment alignment are foundational moves in models from Dick et al., (2014) through Gagné et al. (2005), formalized here as a specification an enterprise system can govern and reuse.

- 1. Performance Need Architecture – defines the instructional problem, the performance gap instruction is meant to close, evidence supporting instruction as the appropriate intervention, and organizational context.
- 2. Learner and Stakeholder Architecture – defines who is affected: learner characteristics, stakeholder expectations, prior knowledge, accessibility needs, technology readiness, and environmental considerations.
- 3. Performance Outcome Architecture – specifies measurable learning outcomes, competencies, performance expectations, and intended transfer of learning.

- 4. Assessment Architecture – defines how learning will be measured, including assessment methods, mastery criteria, evidence requirements, and feedback strategies.
- 5. Learning Science Architecture – identifies the evidence-based learning principles that will guide design decisions, including cognitive load management (Sweller et al., 1998), multimedia design (Mayer, 2021), retrieval practice (Roediger & Karpicke, 2006), spaced learning (Cepeda et al., 2006), scaffolding (Wood et al., 1976), motivation, and other applicable instructional theories, and why they apply.
- 6. Instructional Strategy Architecture – defines instructional approaches, sequencing, facilitation strategies, learner activities, instructional methods, and delivery modalities.
- 7. Content Architecture – establishes approved knowledge sources, content scope, terminology, conceptual organization, content boundaries, and prerequisite relationships.
- 8. Learning Experience Architecture – designs the overall learner journey by organizing engagement, interaction, practice, reflection, collaboration, feedback, and assessment into a coherent learning experience.
- 9. Media Strategy Architecture – defines the instructional purpose and appropriate use of text, graphics, audio, video, simulations, interactive technologies, and other learning media.
- 10. Success Architecture – establishes implementation assumptions, success indicators, evaluation criteria, operational constraints, and performance measures used to determine whether the instructional solution achieves its intended outcomes.

Authority over these ten domains sits with qualified human professionals; AI may organize research, draft alternatives, or flag inconsistencies for review, but does not set outcomes, approve

assessments, or authorize the specification. This boundary is not a formality, it is the mechanism that keeps AI-assisted instructional design accountable to the people who will use it.

## 2.4 AI Governance Engine – The Ten Governance Domains

The AI Governance Specification translates the approved Human Design Specification into enforceable rules for how AI may participate. Its ten domains draw on established AI-governance guidance, including the NIST (2023) Artificial Intelligence Risk Management Framework (AI RMF 1.0) and the OECD AI Principles (Organisation for Economic Co-operation and Development [OECD], 2019), and reflect concerns consistently identified across the broader AI-governance literature (Batoool et al., 2025; Mäntymäki et al., 2022), adapted here specifically to instructional production.

- 1. Governance Objective Architecture – defines the governance purpose, objectives, guiding principles, scope, and expected outcomes for AI participation within the instructional initiative.
- 2. Governance Authority Architecture – defines governance roles, decision authority, accountability, approval responsibilities, escalation pathways, and separation of duties.
- 3. AI Operating Boundary Architecture – defines permitted AI capabilities, restricted activities, prohibited uses, required human approvals, and operational constraints governing AI participation.
- 4. Risk and Compliance Architecture – establishes governance controls addressing instructional, ethical, legal, accessibility, privacy, security, reputational, and organizational risks.

- 5. Data and Privacy Architecture – defines approved data sources, data governance requirements, learner privacy protections, security controls, retention requirements, and acceptable data use practices.
- 6. Transparency and Documentation Architecture – defines documentation requirements, audit trails, traceability expectations, explainability requirements, reporting obligations, and governance records.
- 7. Human Oversight Architecture – establishes required human review points, approval checkpoints, override authority, accountability mechanisms, and human responsibilities throughout the AI-assisted workflow.
- 8. Quality Governance Architecture – defines governance expectations supporting independent quality assurance, validation responsibilities, quality accountability, and coordination with the Quality Intelligence Engine.
- 9. Change Management Architecture – establishes governance processes for policy revisions, exception handling, configuration changes, governance version control, and organizational change management.
- 10. Governance Improvement Architecture – defines how governance effectiveness is monitored, evaluated, measured, and continuously improved across instructional initiatives.

Chapter 5 returns to governance at the organizational level, who sits on a governance committee, how policies are documented, so this section focuses only on what a governance specification must contain for a single instructional initiative.

## 2.5 AI Instructional Production Engine – The Nine Instructional Production Domains

The AI Instructional Production Specification converts approved design and governance decisions into an executable production workflow. Where Sections 2.3 and 2.4 are primarily about decisions, this engine is primarily about coordinated execution, specifically instructional asset production, the domains below function more like workflow stages than independent decision areas (Bolick & Da Silva, 2024; Luo et al., 2025).

- 1. Production Planning – defines production objectives, required instructional assets, production scope, resource requirements, schedules, and operational constraints.
- 2. Production Workflow Orchestration – coordinates production activities, task sequencing, dependencies, AI–human collaboration, review checkpoints, and workflow execution.
- 3. Instructional Content Production – develops instructional content, learner materials, facilitator resources, examples, case studies, and learning activities aligned with the approved Human Design Specification.
- 4. Assessment and Learning Resource Production – develops formative and summative assessments, authentic performance tasks, question banks, rubrics, feedback resources, and supporting instructional materials.
- 5. Media and Multimedia Production – develops storyboards, graphics, presentations, illustrations, animations, audio, video, simulations, and other multimedia learning assets supporting the instructional experience.
- 6. Asset Refinement – revises AI-generated instructional assets through subject-matter expert review, instructional

design review, stakeholder feedback, and iterative improvement before production handoff.

- 7. Production Documentation – documents production activities, AI contributions, prompts, production decisions, version history, asset provenance, and workflow records to ensure organizational traceability and reproducibility.
- 8. Asset Integration – assembles instructional assets into a coherent instructional system by integrating learning content, assessments, media, supporting resources, metadata, and implementation components.
- 9. Production Readiness – verifies that instructional assets, production documentation, and supporting materials are complete, internally consistent, and ready for transfer to the Quality Intelligence Engine for independent validation.

Notice the shape of the list: the first two domains enable production, the middle four produce and refine assets, and the last three prepare those assets for independent quality validation – plan, build, hand off.

This engine is where AI activity is heaviest, drafting, generating variations, assembling media assets, but every AI-generated output remains subject to human review and independent quality validation of the Quality Intelligence Engine. No engine approves its own work; that separation is deliberate and is discussed further in Section 2.7.

## 2.6 Quality Intelligence Engine – The Ten Quality Domains

The Quality Intelligence Specification documents an independent validation of instructional products before implementation — independent in the specific sense that the people or systems who produced the material are not the ones certifying it as ready. This separation of production from validation is a long-standing quality-management principle (Juran & Godfrey, 1999) and aligns with

established instructional-quality frameworks, including Quality Matters (n.d.), OSCQR (n.d.), Universal Design for Learning (CAST, 2018), and the Web Content Accessibility Guidelines (World Wide Web Consortium [W3C], 2018), which this engine's domains are designed to be compatible with.

- 1. Validation Planning – defines validation objectives, scope, quality standards, evaluation criteria, evidence requirements, reviewers, and validation methods.
- 2. Instructional Alignment Validation – verifies that learning outcomes, instructional strategies, learning activities, assessments, and instructional resources remain aligned with the approved Human Design Specification.
- 3. Governance Compliance Validation – verifies that AI participation complied with approved governance policies, operational boundaries, documentation requirements, and human oversight expectations defined by the AI Governance Specification.
- 4. Production Integrity Validation – verifies the completeness, consistency, traceability, version control, and reproducibility of instructional production activities and supporting production records.
- 5. Accessibility and Inclusion Validation – verifies compliance with accessibility standards, Universal Design for Learning principles, usability requirements, and inclusive learning practices across instructional assets. This engine's Accessibility Agent (Section 2.9) drafts the accessibility assets themselves, alt text, captions, transcripts; the engine's independent human reviewer, not the agent, certifies the result, the same production-from-validation separation this engine applies everywhere else.

- 6. Quality Evidence and Metrics – defines the evidence, quality indicators, performance measures, evaluation records, and supporting documentation used to justify validation decisions.
- 7. Risk and Issue Validation – identifies unresolved quality issues, instructional deficiencies, governance concerns, accessibility problems, production risks, and implementation constraints requiring resolution.
- 8. Corrective Actions and Recommendations – specifies required revisions, corrective actions, improvement recommendations, remediation priorities, and follow-up validation activities before implementation.
- 9. Implementation Readiness Validation – determines whether instructional assets satisfy established quality requirements and are ready for implementation, conditional approval, or additional revision.
- 10. Continuous Quality Improvement – captures lessons learned, quality trends, recurring issues, validation findings, and recommendations that strengthen future instructional development and organizational quality practices.

The distinction worth holding onto here is that quality intelligence asks whether a system is ready to implement; it does not ask whether it worked once implemented that second question is instructional evaluation, and it depends on data this engine cannot yet have.

## **2.7 Knowledge Intelligence Engine – The Ten Knowledge Domains**

The Knowledge Intelligence Specification closes the loop: validated experience from one instructional initiative becomes a reusable input to the next. This engine's design draws on organizational-learning scholarship distinguishing information storage from institutional

learning, the difference between an archive and an organization that actually gets better at what it does (Argyris & Schön, 1978; DiBella et al., 1996).

- 1. Knowledge Acquisition – captures validated knowledge, decisions, experiences, evidence, and artifacts generated throughout the Human Design, AI Governance, AI Instructional Production, and Quality Intelligence engines.
- 2. Knowledge Organization – classifies, indexes, structures, and organizes enterprise knowledge to support efficient retrieval, reuse, governance, and long-term knowledge management.
- 3. Knowledge Validation – establishes the human review process for determining the reliability, completeness, credibility, organizational value, and reusability of knowledge before it is approved as enterprise knowledge.
- 4. Knowledge Asset Management – manages reusable instructional templates, design patterns, governance artifacts, production workflows, quality resources, prompt libraries, and other enterprise knowledge assets.
- 5. Pattern and Trend Analysis – identifies recurring practices, emerging trends, performance patterns, organizational opportunities, and systemic issues across multiple instructional initiatives.
- 6. Best Practices and Lessons Learned – documents validated best practices, implementation insights, recurring challenges, lessons learned, and evidence-based recommendations for future instructional work.
- 7. Decision Intelligence – transforms validated organizational knowledge into strategic insights that support future instructional, governance, production, quality, and organizational decision making.

- 8. Innovation and Improvement – identifies opportunities for experimentation, innovation, process optimization, capability development, and continuous improvement throughout the instructional system.
- 9. Knowledge Sharing and Reuse – distributes validated organizational knowledge across individuals, teams, departments, and enterprise systems while promoting the consistent reuse of proven instructional practices and organizational assets.
- 10. Organizational Learning and Capability Development – strengthens enterprise capability by integrating validated knowledge into organizational standards, instructional practices, governance processes, quality systems, and institutional expertise.

Only validated knowledge enters the repository, an unverified impression from one project does not become an organizational assumption. That validation requirement is what keeps this engine from quietly degrading into an unmaintained shared drive.

## 2.8 How the Five Engines Depend on One Another

The five engines are sequential in their primary flow, Human Design into AI Governance into AI Instructional Production into Quality Intelligence into Knowledge Intelligence, but revision is not. A finding at the Quality Intelligence stage routinely sends a project back to Production, and occasionally back to Design itself; a governance policy change can require re-validating assets already in production. Table 2.3 summarizes the most common cross-engine effects so that a change in one place can be traced to what else it is likely to touch.

|                  |                                 |
|------------------|---------------------------------|
| A change here... | ...commonly requires revisiting |
|------------------|---------------------------------|

|                                                  |                                                                        |
|--------------------------------------------------|------------------------------------------------------------------------|
| Instructional problem or learner analysis (HDE)  | Performance outcomes, assessment, strategy, and downstream production  |
| Governance policy or AI operating boundary (AGE) | In-progress production workflows and prior quality validations         |
| Production workflow or asset (AIPE)              | Quality validation, and occasionally the original design specification |
| Quality findings (QIE)                           | Production revisions, and governance if the issue is systemic          |
| Organizational knowledge (KIE)                   | Future Human Design specifications and governance policy updates       |

*Table 2.3. Common cross-engine revision triggers.*

The practical implication is that IDOS should not be implemented as five independent teams handing off finished work in one direction. Whoever owns operational coordination (see Chapter 10) needs visibility across all five engines simultaneously, because the most common failure mode in enterprise instructional systems is not a weak engine, it is a strong engine whose output silently drifts out of sync with what the other four now assume to be true.

A handful of domains appear similar enough across engines that they are worth distinguishing explicitly, since the similarity is in the vocabulary, not in what each domain actually governs. Human Design's Domain 10 (Success Architecture) defines success measures before a single asset is built, what “done well” would even mean for this initiative. Quality Intelligence's Domain 9 (Implementation Readiness Validation) is the later, evidence-based decision about whether those measures were actually met, the implementation decision itself, not the criteria it is judged against. Knowledge Intelligence's Organizational Learning and Capability Development (Domain 10) is neither a criterion nor a single decision; it is whether the organization's capacity to design, govern, produce, and validate is improving across many initiatives, not just this one. Success criteria, a readiness decision, and organizational capability are three different

things that happen to share the word “success” or “improvement” in casual conversation about a project.

A second pair is worth the same treatment. Quality Intelligence's Corrective Actions and Recommendations (Domain 8) names what must change on this specific asset before it can ship – a fix, scoped to one project. Its own Continuous Quality Improvement (Domain 10) is narrower than it sounds: what this one review teaches future initiatives, still anchored to a single validation event. Only when many such findings accumulate, get validated, and get organized does Knowledge Intelligence's engine turn them into the kind of durable, reusable organizational capability Section 2.7 describes. The escalation is real and intentional, fix this asset, note this lesson, build organizational capability from many such lessons, not three names for the same activity.

## 2.9 The Cross-Engine AI Agent Taxonomy

Sections 2.3 through 2.7 described what each engine decides and validates. This section describes who, or, more precisely, what, actually drafts, monitors, analyzes, and executes the AI-facing work inside each engine's authority boundary. Every operational engine coordinates a capability; within that capability, four named Core Agent Roles carry out the specific AI-assisted functions that support it. This is the enterprise architecture that underlies the “AI role” columns introduced narratively in Table 3.2 and the ownership assignments in Table 6.2: this section is where those roles are named and organized formally.

The domains in Sections 2.3–2.7 and the agent roles below answer different questions, and it is worth being precise about which is which before the taxonomy itself: a domain answers what decisions must be made; an agent role answers who, or what AI capability, helps perform the resulting work. The relationship between them is deliberately

many-to-many, not one-to-one, a single agent may contribute to several domains (the Content Development Agent touches both Domain 3 and Domain 4 of the AI Instructional Production Engine, for instance), and a single domain may draw on more than one agent role as needed. This is intentional: forcing a strict one-to-one mapping would collapse a real decision structure into an org chart, and IDOS keeps the two separate on purpose – domains belong to the enterprise decision structure Sections 2.3–2.7 define, agent roles belong to the AI execution layer this section defines, and neither substitutes for the other. A third question remains once both are answered: how does an approved decision actually reach an assigned agent as something it can execute? Section 2.9.1 answers that one.

The taxonomy follows a consistent hierarchy, and each level answers a different question:

- Engine – who owns this enterprise capability?
- Core Agent Role – which AI role performs this category of work?
- Illustrative Specialized Functions – what kinds of tasks can that role perform?
- Typical Outputs – what tangible deliverables does it produce?

### Human Design Engine (HDE)

| Core Agent Role              | Illustrative Specialized Functions                                                                                                                 | Typical Outputs                                                                                      |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Research & Analysis Agent    | Learning needs analysis, learner analysis, audience analysis, task analysis, competency analysis, literature synthesis, benchmarking, gap analysis | Needs assessment reports, learner personas, competency maps, environmental scans, analysis summaries |
| Instructional Planning Agent | Course planning, lesson planning, module sequencing, learning pathway design,                                                                      | Course architecture, curriculum maps, module                                                         |

|                         |                                                                                                                                                                   |                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
|                         | storyboard planning,<br>implementation planning                                                                                                                   | sequence, lesson plans,<br>instructional blueprints                                                 |
| Curriculum Design Agent | Curriculum mapping, learning outcome development, scope and sequence design, competency framework design, credential pathway design                               | Curriculum frameworks, competency maps, program structures, learning outcome matrices               |
| Learning Strategy Agent | Instructional strategy selection, active learning design, collaborative learning design, problem-based learning design, simulation planning, microlearning design | Learning strategies, instructional models, activity recommendations, instructional design rationale |

### AI Instructional Production Engine (AIPE)

| Core Agent Role             | Illustrative Specialized Functions                                                                                                                                 | Typical Outputs                                                                                 |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Content Development Agent   | Learning content writing, scenario development, case study development, discussion prompt generation, learning activity development, facilitator guide development | Instructional content, learner materials, facilitator guides, case studies, learning activities |
| Assessment Design Agent     | Quiz generation, question bank development, rubric development, performance task design, reflection activity development, feedback generation                      | Formative assessments, summative assessments, authentic assessments, rubrics, feedback guides   |
| Media Design Agent          | Storyboard creation, presentation design, infographic development, diagram creation, illustration planning, visual asset design                                    | Storyboards, presentation slides, graphics, diagrams, visual design specifications              |
| Multimedia Production Agent | Video scripting, narration writing, podcast development, voice-over scripting, webinar planning, interactive media planning                                        | Narration scripts, video production plans, podcast scripts, multimedia learning resources       |

## Quality Intelligence Engine (QIE)

| Core Agent Role          | Illustrative Specialized Functions                                                                                            | Typical Outputs                                                                         |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Accessibility Agent      | Alt text generation, captioning, transcript creation, plain-language revision, accessibility review, UDL review               | Accessibility documentation, alt text, captions, transcripts, accessibility reports     |
| Quality Assurance Agent  | Learning objective alignment, assessment alignment, instructional consistency review, content quality review, quality scoring | Quality review reports, alignment reports, quality scorecards, revision recommendations |
| Validation Agent         | SME validation, pilot review, production readiness review, instructional validation, stakeholder review                       | Validation reports, production readiness reports, approval recommendations              |
| Quality Monitoring Agent | Quality metrics monitoring, issue detection, quality trend analysis, continuous quality review                                | Quality dashboards, monitoring reports, quality improvement recommendations             |

## AI Governance Engine (AGE)

| Core Agent Role             | Illustrative Specialized Functions                                                                                              | Typical Outputs                                                            |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Governance Compliance Agent | Governance verification, ethical AI review, responsible AI assessment, governance checklist review, lifecycle compliance review | Governance reports, compliance checklists, AI governance assessments       |
| Policy Review Agent         | Policy alignment review, copyright review, privacy review, data governance review, institutional policy review                  | Policy compliance reports, copyright reviews, privacy assessments          |
| Risk Assessment Agent       | Bias assessment, privacy risk analysis, security risk analysis, hallucination review, academic integrity review                 | Risk registers, risk assessment reports, mitigation recommendations        |
| Documentation Agent         | Decision logging, audit trail generation, AI usage documentation, governance                                                    | Decision logs, audit records, governance documentation, compliance records |

|  |                                  |  |
|--|----------------------------------|--|
|  | reporting, version documentation |  |
|--|----------------------------------|--|

### Knowledge Intelligence Engine (KIE)

| Core Agent Role              | Illustrative Specialized Functions                                                                                 | Typical Outputs                                                                                       |
|------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Learning Analytics Agent     | Learner performance analysis, engagement analysis, completion analysis, outcome evaluation, predictive analysis    | Learning analytics dashboards, performance reports, evaluation summaries                              |
| Knowledge Capture Agent      | Lessons learned capture, best practice documentation, SME knowledge capture, retrospective analysis                | Lessons learned reports, best practice repositories, knowledge summaries                              |
| Repository Management Agent  | Asset organization, metadata tagging, repository indexing, version management, knowledge classification            | Organized repositories, metadata records, asset catalogs, version histories                           |
| Continuous Improvement Agent | Improvement recommendation, process optimization, trend analysis, capability assessment, innovation recommendation | Improvement plans, optimization reports, maturity assessments, continuous improvement recommendations |

*Table 2.4. The twenty core AI agent roles, grouped by operational engine, four roles per engine.*

*Note. The specialized functions listed are illustrative rather than exhaustive. Organizations may implement additional functions within each core agent role, provided they align with the responsibilities of the corresponding engine. This ensures that the IDOS architecture remains stable while accommodating future advances in AI capabilities and evolving organizational needs.*

The separation this table makes explicit is clean, and it maps directly onto the five engines already established in Sections 2.3–2.7:

- The Human Design Engine decides what should be designed; its agents research and draft in support of that decision.

- The AI Instructional Production Engine creates instructional assets; its agents generate the content, assessments, and media a human reviews.
- The Quality Intelligence Engine verifies whether the assets meet enterprise standards; its agents analyze evidence and flag issues, but do not certify readiness themselves.
- The AI Governance Engine ensures the work complies with organizational policies, ethics, and risk controls; its agents monitor and document, they do not set policy.
- The Knowledge Intelligence Engine captures organizational knowledge and drives continuous learning and improvement; its agents identify patterns and organize evidence, while qualified human reviewers approve what becomes reusable organizational knowledge. (Table 2.2).

Within IDOS, an agent is a functional role, not a specific AI platform, a single platform (Claude, ChatGPT, Gemini, Midjourney, Adobe, organization-specific AI agents, multi-agent systems, or a future system) may perform one or more of these twenty roles depending on its capabilities, which is what keeps the framework platform-independent (Principle 8, Section 1.6) even as the underlying technology changes.

### ***2.9.1 Enterprise Prompt Architecture***

Section 2.3–2.7 established what decisions each engine's domains require. Section 2.9 established who, which Core Agent Role, carries out the AI-facing work inside each domain. This section closes the remaining gap: how an approved decision actually reaches an agent as an executable instruction (Chen et al., 2025). The mechanism is the Enterprise Prompt Architecture (EPA).

Within IDOS, a prompt is not a casual conversational request. It is a structured operational instruction that communicates an approved

enterprise decision, drawn from one or more of the five specifications, to an AI agent for a defined instructional task. The EPA is the mechanism that performs this translation: rather than being authored from personal preference or ad hoc phrasing, a prompt is constructed from the same Human Design, AI Governance, AI Instructional Production, Quality Intelligence, and Knowledge Intelligence specifications that govern everything else in this chapter. Stated as this book states its other central commitments: specifications define enterprise decisions; prompts operationalize those decisions for AI execution. The EPA does not replace the five specifications, it is the layer that carries their content forward into an agent's actual instructions.

### **Prompt inheritance**

A prompt inherits context only from the specification, or specifications, its assigned agent role and task actually require, not from all five by default. A single-domain task inherits from one specification; a cross-engine task inherits from several. An Accessibility Agent reviewing a draft, for instance, inherits primarily from the Quality Intelligence Specification. A Content Development Agent drafting new material inherits more broadly: instructional objectives from the Human Design Specification, operating boundaries from the AI Governance Specification, format and workflow requirements from the AI Instructional Production Specification, accessibility expectations from the Quality Intelligence Specification, and approved terminology or templates from the Knowledge Intelligence Specification. This selective inheritance, not automatic access to everything, is what keeps a prompt scoped to the task in front of it rather than accumulating unrelated constraints as the architecture grows. Table 2.5 gives representative examples of what each specification typically contributes.

| Operational engine                 | Example inherited context                                                                                                         |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Human Design Engine                | Performance needs, learner characteristics, learning outcomes, instructional strategy, content architecture, media strategy.      |
| AI Governance Engine               | AI operating boundaries, governance rules, privacy requirements, copyright guidance, risk controls, human oversight requirements. |
| AI Instructional Production Engine | Production objectives, instructional asset type, workflow requirements, required outputs, output formats.                         |
| Quality Intelligence Engine        | Accessibility requirements, quality standards, validation criteria, corrective action requirements.                               |
| Knowledge Intelligence Engine      | Approved templates, organizational terminology, reusable design patterns, validated best practices.                               |

*Table 2.5. Prompt inheritance across the operational engines — representative, not exhaustive, examples of what each specification typically contributes to a prompt.*

### Prompt templates

IDOS distinguishes a prompt template from a prompt (Mao et al., 2025). A template defines the reusable enterprise structure, the required context and operational expectations, from which an individual prompt is instantiated for a specific task; the prompt itself combines that template with task-specific information. Templates, not one-off prompts, are what an organization should treat as the reusable asset: authored by instructional professionals, generated with AI assistance, or built collaboratively, but governed by the approved specifications rather than by whoever happens to be typing.

### Platform independence

The EPA is platform-independent for the same reason the rest of IDOS is (Principle 8, Section 1.6): because a prompt originates from approved specifications rather than from any AI platform's particular interface or capabilities, an organization can change AI models, prompting techniques, or orchestration tools without touching the

Human Design, AI Governance, AI Instructional Production, Quality Intelligence, or Knowledge Intelligence specifications underneath. The specifications are stable; the prompts built on top of them, and the platforms executing those prompts, are expected to change.

### **Prompt lifecycle and governance**

Prompts and prompt templates are enterprise operational assets, not disposable interactions, and are governed accordingly: created, reviewed, validated, version-controlled, revised, and retired as organizational needs change. A template that consistently produces reliable results becomes a candidate for the Knowledge Intelligence Engine's repository (Section 2.7), where it is validated and made reusable rather than left in one person's history of past conversations. Because a prompt operationalizes an enterprise decision, it inherits that decision's governance obligations too, the same version control, documentation, and review expectations Section 2.4 establishes apply here, not a separate, lighter standard.

Read this way, prompt engineering within IDOS is not an individual prompting skill but an enterprise capability: the disciplined practice of translating approved specifications into governed, reusable prompts an agent can execute consistently, transparently, and accountably, wherever in the organization that agent's work is needed.

## **2.10 A Note on Recurring Numbers**

Five engines. Five readiness pillars (Chapter 4). Five governance domains (Chapter 5). A five-way asset classification (Chapter 9). A careful reader is right to ask whether these numbers were discovered independently or imposed for symmetry.

The honest answer is mixed, and worth stating rather than hoping no one notices. The five operational engines are foundational — every other structure in this book is organized around them, and the number

five reflects five genuinely distinct concerns (instructional intent, AI governance, production, validation, organizational knowledge), not a stylistic choice. Where the same number recurs elsewhere, it is sometimes because a five-part structure at comparable granularity is easier for a practitioner to hold in memory alongside the engines, and sometimes coincidental. Where five was not the right number, this book does not force it: the AI Policy Framework has ten domains (Table 5.4), the enterprise workflow has six stages (Table 6.1), the modernization lifecycle has nine (Table 9.4), and each of the four design-object frameworks in Chapter 8 has eight. Even the ten-domain pattern shared by four of the five engines is not treated as sacred: the AI Instructional Production Engine has nine (Section 2.5). If every structure in this book landed on five, or every engine's domain count landed on ten no matter what, that would be a legitimate reason for suspicion. Not every one does, and that variation is the evidence that the numbers follow the content rather than the reverse.

The agent taxonomy introduced in Section 2.9 contains four roles per engine, twenty in total. Not every decision domain requires a corresponding AI agent, however. The Knowledge Intelligence Engine includes a Knowledge Validation domain because determining what becomes approved organizational knowledge is an essential enterprise decision. Final approval of organizational knowledge is intentionally assigned to qualified human reviewers, while AI supports analysis and evidence organization throughout the validation process (Table 2.2).

## 2.11 Chapter Summary

The five operational engines share one architecture, a philosophy, a specification, a set of decision domains, an authority boundary, and a common set of enterprise outputs, applied to five different concerns: instructional intent, AI governance, production, quality, and organizational knowledge. Section 2.9 showed that this same shared

architecture extends one layer further, into a common taxonomy of twenty AI agent roles, four per engine, that execute work within each engine's authority boundary. Section 2.9.1 closed the remaining gap, how an approved decision actually reaches an assigned agent, through the Enterprise Prompt Architecture. Human authority sits at every decision point; AI sits inside a support role that a human reviews before it becomes binding, whether that support comes from a named agent role, a governed prompt, or the more general drafting and monitoring functions described narratively in Chapter 3. Chapter 3 explains how these five engines function together as one integrated system across a complete instructional lifecycle.

## Chapter 3

### The Integrated Operating System

*Chapter 2 examined the five operational engines as a comparative model, along with the agent roles that execute work inside each one. This chapter explains how they function together as a single enterprise system – how information moves between them, how Human–AI collaboration and governance stay consistent across the whole lifecycle rather than engine by engine, and how the framework differs from, and complements, existing instructional-design scholarship. Chapter 4 opens Part II by turning this architecture into an implementation plan.*

#### 3.1 The Operational Objects as a Single Source of Truth

Each engine's authoritative specification, the HDS, AGS, AIPS, QIS, and KIS introduced in Chapter 2, is meant to function as the single source of truth for its domain. Table 3.1 shows how each depends on the one before it, and where each ultimately feeds.

| Object            | Depends on                                    | Feeds into |
|-------------------|-----------------------------------------------|------------|
| HDS (design)      | Enterprise goals, organizational requirements | AGS        |
| AGS (governance)  | HDS                                           | AIPS       |
| AIPS (production) | HDS, AGS                                      | QIS        |
| QIS (quality)     | AIPS, HDS, AGS                                | KIS        |
| KIS (knowledge)   | QIS, implementation experience                | Future HDS |

*Table 3.1. How the five operational objects depend on and feed one another.*

The practical value of insisting on a single authoritative object per engine, rather than letting each team keep its own working notes, is that when a requirement changes, there is exactly one place to update it, and every downstream engine is working from the same version

rather than from conflicting assumptions. Without this discipline, the most common failure in AI-assisted instructional programs is not bad instructional design or ungoverned AI use in isolation; it is quiet drift between what different parts of the organization believe was agreed.

### 3.2 Human-AI Collaboration Across the System

Chapter 2 established that authority sits with people at every engine while AI occupies a support role, carried out by the named agent roles introduced in Section 2.9 and reaching those agents as governed prompts built from the specifications themselves (Section 2.9.1). That pattern holds consistently across the lifecycle, but the specific kind of support AI offers shifts as work moves from design toward implementation (Table 3.2).

| Stage             | Typical human role                            | Typical AI role                                            |
|-------------------|-----------------------------------------------|------------------------------------------------------------|
| Design (HDE)      | Define outcomes, strategy, assessment         | Organize research, draft alternatives for review           |
| Governance (AGE)  | Set policy, approve exceptions                | Monitor compliance, flag risk                              |
| Production (AIPE) | Review, refine, approve                       | Draft content, generate variations, automate routine tasks |
| Quality (QIE)     | Independent validation decision               | Summarize evidence, detect inconsistencies                 |
| Knowledge (KIE)   | Approve what becomes organizational knowledge | Identify patterns across many projects                     |

*Table 3.2. How the balance of Human-AI collaboration shifts across the instructional lifecycle. Each “Typical AI role” is carried out by one or more of the four Core Agent Roles assigned to that engine in Section 2.9.*

This is a design commitment, not an incidental detail: the Human-in-Governance principle holds that AI may inform decisions anywhere in the system, but it does not govern the system. Organizations that blur

this line, letting an AI-drafted recommendation quietly become the de facto decision because no one formally overrode it, tend to lose the accountability IDOS is built to preserve, even while every individual step looked reasonable in isolation.

### 3.3 Governance Is Distributed, Not Centralized in One Engine

It would be a natural misreading of Chapter 2 to conclude that only the AI Governance Engine does governance work. In practice, each engine applies governance appropriate to its own function, the Human Design Engine governs instructional decision quality, the AI Instructional Production Engine governs production consistency, the Quality Intelligence Engine governs validation independence, and the Knowledge Intelligence Engine governs what counts as reliable organizational knowledge. Chapter 5 develops the organizational structures (committees, roles, decision rights) that coordinate this distributed governance; the point to establish here is architectural: governance is a property of the whole system, not a department bolted onto it.

### 3.4 Why IDOS Is Different — Revisited

Section 1.7 compared IDOS to conventional instructional design in outline form. With the full architecture now in view, three distinguishing claims can be stated more precisely.

- IDOS does not compete with ADDIE, Backward Design, SAM, or Agile instructional design, it governs the enterprise environment in which any of them operates once AI is involved (Molenda, 2003; Wiggins & McTighe, 2005; Allen & Sites, 2012).
- IDOS treats AI governance as inseparable from instructional design, rather than as a compliance layer added after content

already exists, governance decisions in Chapter 2's AGE precede, not follow, production.

- IDOS treats organizational learning as an operating requirement, not a retrospective activity, the Knowledge Intelligence Engine is a standing part of the lifecycle, not an end-of-project survey.

These are architectural claims about how the framework is designed, not empirical claims about outcomes it has been shown to produce. No implementation of IDOS has yet been evaluated in published, peer-reviewed research, and the claims in this section should be read as the reasoning behind the framework's design rather than as demonstrated results. The back matter's "Limitations and Future Research" section addresses this directly and is intended to be read alongside this chapter, not skipped.

### 3.5 Chapter Summary

The five operational objects form one continuous chain of evidence, each depending on the one before it and feeding the one after. Human-AI collaboration follows a consistent authority pattern across the lifecycle, carried out by the agent roles introduced in Section 2.9, even as the specific nature of AI's support shifts from drafting to monitoring to pattern analysis. Governance is distributed across all five engines rather than confined to one. And the framework's distinguishing claims, methodology-neutrality, governance-first design, and standing organizational learning, are architectural commitments whose value has not yet been independently validated at scale. Part II turns from this integrated architecture to the practical work of implementing it.

PART II

**Implementation & Governance**

## Chapter 4

### Enterprise Readiness and Implementation Planning

*Understanding IDOS and implementing it are different problems. This chapter presents a structured way to assess an organization's readiness before committing to implementation, prioritize what to fix first, and sequence a realistic rollout. The fillable Enterprise Readiness Assessment (ERA) template that operationalizes this chapter is in Appendix A.1; this chapter explains the reasoning behind it rather than repeating its fields.*

#### 4.1 Why Readiness Assessment Comes First

Organizations that begin AI-assisted instructional work without assessing readiness tend to discover the same problems repeatedly: unclear ownership, inconsistent AI use across teams, and governance built retroactively after a problem has already occurred. A readiness assessment is not a gate that says an organization may or may not proceed, it is a diagnostic that tells leaders which of the five pillars below need attention before scale, and which can be strengthened in parallel with an early pilot.

#### 4.2 The Five Readiness Pillars

Readiness is not one score (Weiner, 2009). It is evaluated across five interdependent pillars, because strength in one cannot substitute for weakness in another, strong technology cannot compensate for absent governance, and enthusiastic leadership cannot substitute for a workforce that has not been prepared for the change (Table 4.1).

| Pillar | Central question |
|--------|------------------|
|--------|------------------|

|            |                                                                                   |
|------------|-----------------------------------------------------------------------------------|
| Leadership | Are leaders visibly committed, and have they allocated real resources?            |
| Governance | Do policies, decision authority, and oversight already exist in some form?        |
| Workforce  | Do instructional teams understand AI-assisted work and their role in it?          |
| Operations | Are current instructional workflows documented and consistent enough to build on? |
| Technology | Can current systems support governed AI-assisted production?                      |

*Table 4.1. The five enterprise readiness pillars.*

These five pillars are not the five operational engines from Chapter 2, and the shared number is coincidental rather than a hidden mapping, Section 2.10 addresses this directly. Readiness asks whether the organization is prepared to run any of the five engines at all; the engines themselves are what readiness is preparing the organization for. A useful way to hold the two apart: an organization can be fully ready, in this chapter's sense, before it has produced a single Human Design Specification, and a mature Knowledge Intelligence Engine (Chapter 2, Section 2.7) is itself evidence of high organizational readiness, not a readiness pillar in its own right.

The Technology pillar's central question in Table 4.1, whether current systems can support governed AI-assisted production, is easy to answer too quickly by checking infrastructure alone. Technology receives closer treatment in Table 4.1a, as the other four pillars already have a dedicated section elsewhere in this book, Governance in Chapter 5, Operations in Chapter 6, Leadership and Workforce woven through Chapters 4 through 7. The Technology pillar's question also requires evaluating the AI tools themselves. Chapter 2 governs how an adopted tool may be used, but not how an organization should decide which tool to adopt for a given agent role in the first place. Table 4.1a gives the criteria that decision should rest on, evaluated per candidate

tool and per intended agent role (Section 2.9), not once for the organization as a whole — a tool well suited to a Research & Analysis Agent's work is not automatically suited to an Accessibility Agent's unless it was designed for both function. The criteria are consistent with, not separate from, established AI risk-management guidance (NIST, 2023), which treats third-party and vendor risk as part of AI risk management rather than an afterthought to it.

| Evaluation criterion            | What it establishes                                                                                                                                                                                                                                                       |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Capability fit                  | Whether the tool's actual outputs, not vendor claims, meet the specific agent role's task requirements at an acceptable error rate.                                                                                                                                       |
| Bias and fairness               | Whether the tool's outputs vary in quality or appropriateness across different learner populations, distinct from overall capability, and consistent with the equity and critical-evaluation expectations Chapter 11 names as a practitioner competency in its own right. |
| Data handling and privacy       | Where learner or institutional data goes if entered into the tool, whether it is used for further model training, and whether that meets the AI Governance Specification's Data and Privacy Architecture (Section 2.4, Domain 4).                                         |
| Security                        | Authentication, access control, and incident response the vendor provides for the tool itself, distinct from what happens to the data entered into it, and treated as its own AI Policy Framework domain (Table 5.4) for that reason.                                     |
| Transparency and explainability | What the vendor discloses about training data, known limitations, and failure modes — the input this organization's own Transparency and Documentation Architecture (Section 2.4, Domain 6) depends on being able to trust.                                               |
| Accessibility of outputs        | Whether the tool's outputs meet, or can be made to meet, the accessibility standards Quality Intelligence validates against (Section 2.6, Domain 5).                                                                                                                      |
| Cost and sustainability         | Total cost at the scale the organization actually intends to operate at, not pilot-scale pricing, and the vendor's likely viability over the specification's useful life.                                                                                                 |

|                       |                                                                                                                                                              |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Vendor accountability | Whether the vendor provides an audit trail, incident disclosure process, and support channel adequate to the risk level of the content this tool will touch. |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|

*Table 4.1a. AI tool evaluation criteria, applied per candidate tool and per intended agent role before adoption.*

Formal re-evaluation against these eight criteria happens once per tool per role, not once per project: a tool approved for the Content Development Agent's work does not need re-evaluating every time that role is used. But not re-evaluating by default is not the same as not watching — an organization only knows a tool, a role's requirements, or a vendor's practices have changed if something is monitoring for that change between evaluations. That ongoing monitoring is not a new obligation this table introduces; it is the AI Policy Framework's existing Continuous improvement domain (Table 5.4), which already defines expectations for monitoring AI performance and updating policy, applied here to the specific question of when a re-evaluation is due. The Enterprise Readiness Assessment template (Appendix A.1) records the evaluation outcome under the Technology pillar; the AI Governance Specification (Section 2.4) is where an approved tool's operating boundaries, not its initial suitability, get defined.

4.3 The Readiness Maturity Model

Each pillar is scored against a common five-level maturity model, which allows an organization to be strong in one pillar and weak in another without forcing a single overall verdict (Table 4.2). The five-level structure, and the Initial-to-Optimizing progression in particular, follows the pattern established by the Capability Maturity Model (Paulk et al., 1993), adapted here to instructional readiness rather than software process maturity. Consistent with established AI-governance guidance (NIST, 2023) and quality-management practice (International Organization for Standardization [ISO], 2023), maturity is treated as a

continuum an organization moves along, not a binary readiness test it passes or fails.

| Level          | Description                                                             |
|----------------|-------------------------------------------------------------------------|
| 1. Initial     | Activities are informal, inconsistent, or largely absent.               |
| 2. Emerging    | Basic structures exist but are applied inconsistently.                  |
| 3. Established | Core capabilities are documented, repeatable, and consistently applied. |
| 4. Integrated  | Activities are coordinated across the organization and governed.        |
| 5. Optimizing  | Continuously evaluated and improved through organizational learning.    |

*Table 4.2. The five-level enterprise readiness maturity model, applied independently to each pillar.*

As a rough guide: organizations scoring mostly at Levels 1–2 should strengthen foundational capability before attempting anything beyond a tightly governed pilot; Level 3 organizations are generally ready for phased implementation; Levels 4–5 are positioned to expand across departments while continuing to refine. These are starting heuristics, not fixed rules, the right pace depends on organizational risk tolerance and the stakes of the instructional content involved.

## 4.4 Prioritizing Capability Gaps

A readiness assessment typically surfaces more gaps than an organization can address at once. Not every gap deserves the same urgency, so IDOS classifies them into three priority levels (Table 4.3) rather than treating every finding as equally blocking.

| Priority | Meaning                                 | Action                                    |
|----------|-----------------------------------------|-------------------------------------------|
| Critical | Creates significant implementation risk | Resolve before large-scale implementation |

|               |                                  |                                                |
|---------------|----------------------------------|------------------------------------------------|
| Important     | Affects quality but not blocking | Address during phased implementation           |
| Developmental | Long-term capability opportunity | Address through ongoing continuous improvement |

*Table 4.3. Capability gap priority levels.*

A governance policy that does not yet exist at all is typically critical; a workforce that is only moderately confident using AI tools is typically important, addressable through professional learning delivered in parallel with early implementation; a knowledge-sharing practice that is informal but functional is typically developmental. The full gap-analysis template, including root-cause prompts and a capability gap register, is provided in Appendix A.1.

## 4.5 The Four-Phase Implementation Roadmap

IDOS recommends implementing in four progressive phases rather than attempting enterprise-wide deployment at once (Table 4.4). Each phase builds the evidence and confidence needed for the next.

| Phase       | Primary goal                                                  | Typical duration |
|-------------|---------------------------------------------------------------|------------------|
| 1. Prepare  | Establish governance, communicate goals, complete the ERA     | 2–8 weeks        |
| 2. Pilot    | Test the framework within one team or program                 | 2–6 months       |
| 3. Expand   | Scale to additional departments or the full enterprise        | Varies by scope  |
| 4. Optimize | Strengthen governance, workflows, and capability continuously | Ongoing          |

*Table 4.4. The four-phase enterprise implementation roadmap.*

The pilot phase deserves particular emphasis: it is where an organization discovers whether its governance policies actually work under real conditions, before those policies are asked to scale. Skipping

directly to expansion is the single most common implementation mistake IDOS is designed to prevent, not because pilots are bureaucratic, but because governance assumptions that look sound on paper reliably surface gaps once real instructional teams and real AI tools are involved.

***Field Note (illustrative)***

*A mid-size university's teaching-and-learning center piloted IDOS with a single online course redesign before touching anything else. The Prepare phase took six weeks, mostly spent discovering that “who approves AI-generated assessment items” had never actually been decided, only assumed. That single unresolved question, surfaced early and cheaply during a pilot, would have stalled a department-wide rollout for months if it had surfaced there instead.*

## 4.6 Common Implementation Risks

Table 4.5 summarizes the risks that recur most often across organizations implementing AI-assisted instructional systems (Rane et al., 2024; McNeill et al., 2025), along with the response that most directly addresses each. None of these indicate that implementation has failed, they are ordinary friction points an implementation plan should anticipate rather than be surprised by.

| Risk                               | Typical response                                                                    |
|------------------------------------|-------------------------------------------------------------------------------------|
| Limited leadership support         | Name an executive sponsor; communicate a clear vision and priorities                |
| Unclear or absent governance       | Establish policies and decision authority before expanding AI use                   |
| Workforce resistance               | Involve staff early; provide practical, role-specific professional learning         |
| Inconsistent operational processes | Document current workflows before standardizing; allow reasonable local flexibility |
| Technology limitations             | Assess infrastructure before committing to a rollout timeline; adopt incrementally  |

|                                 |                                                                  |
|---------------------------------|------------------------------------------------------------------|
| Limited organizational learning | Document lessons learned as a required step, not an optional one |
|---------------------------------|------------------------------------------------------------------|

*Table 4.5. Common enterprise implementation risks and typical responses.*

## 4.7 Chapter Summary

Readiness is assessed across five pillars, leadership, governance, workforce, operations, and technology, using a shared five-level maturity model, so that strengths in one area cannot mask weaknesses in another. Capability gaps are triaged by priority rather than treated uniformly, and implementation proceeds through four phases, Prepare, Pilot, Expand, Optimize, rather than as a single enterprise-wide launch. The Enterprise Readiness Assessment template in Appendix A.1 operationalizes this chapter into a document leaders can complete directly. Chapter 5 turns to the governance structures that a readiness assessment typically identifies as most urgently needed.

## Chapter 5

### Enterprise Governance and Operating Model

*Chapter 2 introduced governance as one of the five operational engines, focused on a single instructional initiative. This chapter addresses governance at the organizational level – the standing committees, roles, and decision rights that operate across every initiative once IDOS is in production use. The fillable Enterprise Governance Plan (EGP) template is in Appendix A.2; this chapter explains the reasoning a leadership team needs before completing it.*

#### 5.1 From Project Governance to Organizational Governance

A single instructional project can be governed informally by a small, engaged team. An organization running many AI-assisted instructional initiatives across departments cannot, without a standing structure, different teams will use AI inconsistently, apply different quality standards, and duplicate governance decisions that should only need to be made once. Organizational governance exists to prevent that fragmentation while remaining lightweight enough that it does not become a bottleneck instructional teams route around (Müller, 2016).

#### 5.2 The Five Governance Domains

Organizational governance is distributed across five domains, each with a distinct focus (Table 5.1), a structured, domain-based approach consistent with layered AI-governance models proposed elsewhere in the literature (Agarwal & Nene, 2025). Organizations should assign these domains to existing leaders or committees wherever possible rather than creating new structures from scratch, what matters is that every domain has a clear owner, not that a new department is built to hold it.

| Domain                 | Primary focus                            | Primary outcome                    |
|------------------------|------------------------------------------|------------------------------------|
| Leadership governance  | Strategic direction and resourcing       | Organizational alignment           |
| Operational governance | Daily coordination of instructional work | Consistent execution               |
| AI governance          | Responsible AI use and policy compliance | Governed, ethical AI participation |
| Quality governance     | Enterprise-wide quality standards        | Verified instructional quality     |
| Knowledge governance   | Organizational learning and reuse        | Enterprise intelligence            |

*Table 5.1. The five organizational governance domains.*

Three of these five domains map fairly directly onto engines from Chapter 2: AI governance oversees the AI Governance Engine's work across every initiative, quality governance oversees the Quality Intelligence Engine's, and knowledge governance oversees the Knowledge Intelligence Engine's. Leadership governance and operational governance do not map to a single engine the same way, they sit above all five, covering the strategic resourcing and daily coordination (Section 2.8's cross-engine drift, at organizational scale) that no individual engine is responsible for on its own. The two structures are related, not identical, and organizations should not expect a clean one-to-one org chart between them.

### 5.3 Decision Authority

The most common governance failure is not the absence of policy, it is ambiguity about who has authority to make a given decision, which causes decisions to stall or to be made informally by whoever is most available. Table 5.2 gives a starting decision-authority matrix; organizations should adapt it to their own leadership structure rather

than adopt it verbatim, but every decision category should have exactly one primary owner.

| Decision area                                  | Typical primary authority                           |
|------------------------------------------------|-----------------------------------------------------|
| Enterprise strategy and resourcing             | Executive leadership                                |
| Governance policy updates                      | Governance lead, in consultation with domain owners |
| Daily instructional operations                 | Operational leadership                              |
| AI tool approval and risk decisions            | AI governance lead                                  |
| Quality standards and implementation readiness | Quality governance lead (independent of production) |
| What becomes reusable organizational knowledge | Knowledge governance lead                           |

*Table 5.2. Starting decision-authority matrix; adapt to organizational structure.*

Two design choices are worth calling out explicitly. First, the quality governance lead's authority is deliberately kept independent of the teams producing instructional content, the same separation-of-production-from-validation principle introduced in Chapter 2's Quality Intelligence Engine, now applied at the organizational level. Second, routine operational decisions should not require executive sign-off; escalation is reserved for decisions with real strategic, legal, or reputational consequence. Over-escalating routine choices is as damaging to a governance system's credibility as under-governing genuinely risky ones.

***Field Note (illustrative)***

*A corporate learning team spent three weeks debating whether every AI-drafted quiz question needed director-level sign-off, because no decision-authority matrix existed and everyone defaulted to escalating upward to be safe. Once a matrix like Table 5.2 assigned routine assessment review to the instructional design lead and reserved director sign-off for content involving legal or safety claims, the same review cycle dropped to under two days —*

*not because the content got less scrutiny, but because the right level of scrutiny was applied the first time.*

### 5.4 Governance Documentation

Governance operates through four key kinds of documentation, distinguished by how often each should change (Table 5.3) (Agarwal & Nene, 2025; Winecoff & Bogen, 2025). Conflating these, for example, writing operational how-to guidance into a policy document, tends to produce documentation that is either too rigid to be practical or too informal to be enforceable.

| Type                 | Primary purpose                           | Answers                                                                                      | Typical review cycle                |
|----------------------|-------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------|
| Policy               | Defines organizational requirements       | What must the organization do?                                                               | Annual, or when regulations change  |
| Standard             | Defines measurable expectations           | What level of performance is expected?                                                       | Annual                              |
| Procedure            | Describe repeatable processes             | How is the work performed?                                                                   | As processes change                 |
| Operational guidance | Provides practical implementation support | What recommendations, examples, and best practices help people perform the work effectively? | Continuously, as lessons accumulate |

*Table 5.3. The four governance documentation types.*

### 5.5 Enterprise AI Policy Framework

Section 5.4 distinguishes policies, standards, procedures, and operational guidance without specifying what an enterprise AI policy framework should include. Although all four documentation types are essential, policies establish the governing requirements from which standards, procedures, and operational guidance are derived.

Accordingly, Table 5.4 presents a synthesis of ten policy domains that together constitute a comprehensive enterprise AI policy framework for AI-assisted instructional work. The framework draws on established AI governance guidance, including the NIST (2023) Artificial Intelligence Risk Management Framework (AI RMF 1.0), ISO/IEC 42001:2023 Artificial Intelligence–Management System, the OECD (2019) Recommendation of the Council on Artificial Intelligence, and the Privacy Commissioner for Personal Data's [PCPD] (2025) Guidance on the Ethical Development and Use of Artificial Intelligence. Not every organization needs a separate policy document for each domain; smaller organizations may combine multiple domains into a single policy. However, every domain should be addressed somewhere in the organization's governance documentation before AI-assisted instructional work scales beyond a single pilot.

| Policy domain                       | Purpose                                                                                                                                   |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Acceptable use                      | Defines permitted (Green), restricted or approval-required (Amber), and prohibited (Red) AI use in instructional and administrative work. |
| Human oversight                     | Specifies which activities require human review before implementation                                                                     |
| Transparency and disclosure         | Defines when and how AI involvement is communicated to learners and stakeholders                                                          |
| Data privacy and protection         | Governs collection, use, storage, and protection of learner and institutional data                                                        |
| Intellectual property and copyright | Defines ownership, attribution, and licensing of AI-assisted materials                                                                    |
| Accessibility and inclusion         | Ensures AI-supported resources meet accessibility requirements                                                                            |
| Security                            | Establishes authentication, access control, and incident response for AI systems                                                          |
| Quality assurance                   | Defines standards for reviewing and approving AI-generated outputs                                                                        |

|                              |                                                                          |
|------------------------------|--------------------------------------------------------------------------|
| Risk and incident management | Establishes procedures for reporting and mitigating AI-related incidents |
| Continuous improvement       | Defines expectations for monitoring AI performance and updating policy   |

*Table 5.4. The ten AI policy domains that together constitute an enterprise AI policy framework.*

These domains apply uniformly across every context in this book where AI participates, new design (Chapter 8), modernization (Chapter 9), and the case studies in Chapters 10 and 11 all operate within the same ten-domain policy set rather than each requiring its own governance framework. Policies within each domain should be clear, consistent with one another, actionable enough to guide everyday decisions, reviewed periodically as AI capabilities and regulations change, and communicated to everyone doing AI-assisted instructional work, a policy nobody has read governs nothing.

## 5.6 Monitoring Governance Effectiveness

Governance should be judged by outcomes, not by activity, the number of meetings held or policies signed says little on its own (Rotberg, 2014). A governance system is working when it visibly improves decision-making, instructional quality, and responsible AI use; it is not working when instructional teams route around it because it is too slow, or when policies exist on paper but are not reflected in daily practice. Chapter 7 develops enterprise performance measurement in full, including specific indicators for governance effectiveness; this chapter's role is to ensure a governance structure exists that Chapter 7 will have something meaningful to measure.

## 5.7 Chapter Summary

Organizational governance distributes responsibility across five domains, leadership, operations, AI, quality, and knowledge, each with

a clear decision owner, documented through policies, standards, procedures, and operational guidance that are reviewed on different cycles appropriate to how often each should reasonably change, with AI policy specifically organized around the ten domains in Table 5.4. Quality governance is kept structurally independent of production, mirroring the separation established in Chapter 2. The Enterprise Governance Plan template in Appendix A.2 turns this chapter into a document a governance committee can adopt directly. Chapter 6 turns from governance structure to daily operational execution.

## Chapter 6

### Operationalizing IDOS

*Readiness (Chapter 4) and governance (Chapter 5) establish the conditions for implementation. This chapter addresses daily execution: the workflow architecture that coordinates instructional teams and AI across a project, and how that execution is monitored and improved over time. Human–AI collaboration patterns were established once in Section 3.2 and are not restated here – this chapter applies them to a concrete workflow rather than re-describing them. The Operational Execution Framework (OEF) template is in Appendix A.3.*

#### 6.1 The Six-Stage Enterprise Workflow

IDOS does not prescribe a specific instructional-design methodology, organizations may continue using ADDIE, Backward Design, SAM, or an Agile approach within the Human Design Engine (Section 1.7). What IDOS adds is a consistent enterprise workflow architecture around whatever methodology a team uses, so that governance, Human–AI collaboration, and quality checkpoints are applied consistently regardless of instructional approach. That architecture has six stages (Table 6.1).

| Stage       | Primary activity                                            | Typical governance checkpoint            |
|-------------|-------------------------------------------------------------|------------------------------------------|
| 1. Plan     | Define scope, assign roles, review governance requirements  | Governance approval before design begins |
| 2. Design   | Define outcomes, strategy, assessment (Human Design Engine) | Instructional design review              |
| 3. Develop  | Produce assets, with AI support under governance boundaries | AI output verification                   |
| 4. Validate | Independent quality and accessibility review                | Implementation-readiness decision        |

|              |                                                       |                                    |
|--------------|-------------------------------------------------------|------------------------------------|
| 5. Implement | Deploy, support learners, monitor rollout             | Operational readiness confirmation |
| 6. Learn     | Capture lessons learned into organizational knowledge | Knowledge governance review        |

*Table 6.1. The six-stage enterprise instructional workflow.*



Governance checkpoints occur at every stage transition (Table 6.1); the sequence is fixed, the cadence is not

*Figure 6.1. The six-stage workflow. Governance checkpoints (right column of Table 6.1) sit at every transition; how long each stage takes is not fixed — only the sequence and the checkpoints are.*

These six stages map directly onto the five operational engines from Chapter 2: Plan and Design correspond to the Human Design Engine, Develop to the AI Instructional Production Engine (under boundaries set by the AI Governance Engine), Validate to the Quality Intelligence Engine, and Learn to the Knowledge Intelligence Engine. The workflow is the operational expression of the architecture, not a separate model layered on top of it.

## 6.2 Designing Workflows That Apply These Principles Rather Than Copying a Template

Organizations frequently ask whether every instructional project must follow an identical workflow. It should not — what should be consistent is that every workflow, however it is sequenced, embeds governance checkpoints, defines where AI contributes and where a human reviews that contribution, and ends with a knowledge-capture step rather than simply concluding at implementation. A team using Agile sprints can embed governance review into sprint planning; a team using ADDIE can embed it at each phase transition. The six

stages in Table 6.1 describe what must happen, not the order or cadence in which every organization must perform it.

### 6.3 Coordinating the Five Engines in Daily Operation

Section 2.8 described how a change in one engine's output commonly requires revisiting another. Operationally, this means a single team member or small group should hold visibility across all five engines for a given initiative, not because one person should perform all five roles, but because the most common operational failure is not a weak engine, it is a strong engine whose output has silently drifted out of sync with what the other four now assume. Table 6.2 assigns typical operational ownership; organizations should adapt names, not the underlying principle that every engine needs a visible owner.

| Engine                      | Typical operational owner               |
|-----------------------------|-----------------------------------------|
| Human Design                | Instructional design lead               |
| AI Governance               | AI governance lead (Chapter 5)          |
| AI Instructional Production | Production or development lead          |
| Quality Intelligence        | Quality lead, independent of production |
| Knowledge Intelligence      | Knowledge management lead               |

*Table 6.2. Typical operational ownership of the five engines.*

#### ***Field Note (illustrative)***

*A workforce-development team discovered mid-project that their AI Instructional Production lead had quietly started reusing a prompt template the AI Governance lead had flagged for revision two weeks earlier — not from carelessness, but because no one owned watching both engines at once. The fix wasn't more process; it was naming one person whose explicit job was to notice when one engine's output stopped matching what another engine now assumed, exactly the drift Table 2.3 describes.*

## 6.4 Monitoring Operational Execution

Operational monitoring should answer a small number of questions leaders actually need answered, not maximize the number of metrics collected: is work progressing through the six stages without unexplained delay, are governance checkpoints being completed rather than skipped, and is the Learn stage actually producing knowledge that gets reused rather than filed and forgotten? An operational dashboard organized around these questions is more useful than one built by listing every measurable data point available. Chapter 7 develops enterprise performance measurement in full; this section establishes only the operational-level monitoring a project team needs day to day.

## 6.5 Continuous Operational Adaptation

Workflows should be revisited as an organization gains experience — a checkpoint that made sense during an early pilot may become unnecessary friction once a governance policy has proven reliable, and a step that seemed adequate at small scale may need to be more rigorous once AI capabilities or organizational risk exposure change. Treating the six-stage workflow as fixed indefinitely defeats the purpose of Chapter 2's Knowledge Intelligence Engine, whose entire function is to feed operational lessons back into how future work is planned and executed.

## 6.6 Chapter Summary

Daily execution under IDOS follows six stages, Plan, Design, Develop, Validate, Implement, Learn, that map directly onto the five operational engines from Chapter 2, embedding governance checkpoints and Human-AI collaboration into the workflow rather than treating them as separate concerns. Every engine needs a visible operational owner so that drift between engines is caught early. Monitoring should stay focused on whether work is progressing, checkpoints are being

honored, and lessons learned are actually being reused, not on maximizing the number of metrics tracked. The Operational Execution Framework template in Appendix A.3 turns this chapter into a workflow blueprint a project team can adapt directly. Chapter 7 addresses how organizations measure whether all of this is actually working.

## Chapter 7

# Enterprise Performance Measurement and Continuous Improvement

*Readiness, governance, and operations establish how IDOS should run. This chapter addresses how an organization knows whether it is actually working, across governance, operations, Human-AI collaboration, quality, and organizational learning, not just whether individual instructional projects were completed on time. The fillable Enterprise Performance Management Plan (EPMP) template is in Appendix A.4.*

## 7.1 Why Measure the System, Not Just the Projects

An organization can deliver a series of individually successful instructional projects while its underlying operating system quietly deteriorates, governance decisions made inconsistently, AI used differently team to team, lessons learned in one project never reaching the next. Project-level success metrics (completion rates, learner satisfaction) do not surface this. Enterprise performance measurement asks a different, harder question: is the organization's capacity to design, govern, produce, validate, and improve AI-assisted instruction actually growing project over project, or merely repeating itself?

## 7.2 The Six Performance Domains

Performance is evaluated across six domains rather than a single composite score, because the domains can move independently, an organization can improve instructional quality while its governance responsiveness stagnates, or vice versa (Table 7.1).

| Domain | Central question |
|--------|------------------|
|--------|------------------|

|                         |                                                                             |
|-------------------------|-----------------------------------------------------------------------------|
| Strategic performance   | Is IDOS advancing actual organizational goals?                              |
| Governance performance  | Is governance improving decisions, or just documenting them?                |
| Operational performance | Are workflows efficient and well coordinated across engines?                |
| Human–AI collaboration  | Is the collaboration pattern from Section 3.2 actually holding in practice? |
| Quality performance     | Is instructional quality improving and consistent across teams?             |
| Knowledge performance   | Is organizational learning being captured and reused, not just stored?      |

*Table 7.1. The six enterprise performance domains.*

### 7.3 Evaluating the Five Operational Engines

Within those six domains, each of the five operational engines from Chapter 2 can be evaluated using one common model: capability (does it have what it needs to function), effectiveness (is it achieving its purpose), efficiency (is it using resources well), quality (are its outputs reliable), integration (does it coordinate with the other four engines), and continuous improvement (does it contribute to organizational learning). Applying one shared model across all five engines, rather than a different evaluation approach per engine, is what allows leaders to compare engine performance directly instead of comparing incompatible reports.

A caution worth stating plainly: strong performance in any single engine does not indicate a high-performing system. An efficient AI Instructional Production Engine cannot compensate for weak governance; excellent instructional quality does not offset poor knowledge reuse. Evaluate integration, whether the engines are staying in sync, with the same seriousness as evaluating any one engine in isolation.

## 7.4 Indicators, Dashboards, and Decision Support

Every performance indicator an organization tracks should trace back to a specific decision it is meant to inform, governance policy revision, resource reallocation, professional-learning priorities, rather than being collected because it happens to be measurable. Table 7.2 gives a starting set of indicator categories; the full Enterprise Performance Indicator Catalog, including evidence sources and review schedules, is in Appendix A.4.

| Performance domain     | Example indicator                                       | Typical evidence source             |
|------------------------|---------------------------------------------------------|-------------------------------------|
| Governance             | Time from policy exception request to decision          | Governance decision log             |
| Operational            | Workflow stage completion consistency (Ch. 6)           | Operational dashboard               |
| Human-AI collaboration | Proportion of AI outputs requiring major human revision | Production records                  |
| Quality                | Implementation-readiness pass rate on first review      | Quality Intelligence Specifications |
| Knowledge              | Reuse rate of prior knowledge assets in new projects    | Knowledge repository                |

*Table 7.2. Sample enterprise performance indicators by domain.*

A small number of indicators that leaders actually review and act on is more valuable than an exhaustive dashboard no one has time to interpret. Dashboards organized by audience, executive, governance, operational, quality, knowledge, tend to be used more consistently than one dashboard attempting to serve every audience at once.

## 7.5 From Measurement to Organizational Learning

Performance measurement only creates value once findings change what the organization does next. This closes the loop back to Chapter

2's Knowledge Intelligence Engine: a governance finding should update policy, an operational finding should refine the workflow in Chapter 6, and a quality finding should feed back into the Human Design Engine's next specification. An organization that reviews performance data regularly but rarely changes anything as a result has built a reporting habit, not a learning one.

## 7.6 Enterprise Performance Reviews

Performance across all six domains should be reviewed on a regular cadence, dashboards monitored monthly, governance and operational performance reviewed quarterly, and a comprehensive review across all six domains conducted at least annually, feeding directly into the next cycle of readiness assessment (Chapter 4) and governance refinement (Chapter 5). This is what makes IDOS a continuous operating system rather than a one-time implementation project: each annual review becomes the starting evidence for the following year's readiness and governance decisions, rather than a report that is filed and not revisited.

## 7.7 Chapter Summary

Enterprise performance is measured across six domains, strategic, governance, operational, Human-AI collaboration, quality, and knowledge, using one shared evaluation model applied consistently across all five operational engines from Chapter 2. Indicators should trace to specific decisions, not simply be collected because they are measurable, and performance findings should feed back into governance and operational refinement rather than accumulate as unused reports. The EPMP template in Appendix A.4 turns this chapter into a working measurement plan. This closes Part II; Part III applies the complete framework to designing new learning systems and modernizing existing ones.

PART III

**Designing and Modernizing Learning  
Systems**

## Chapter 8

### Enterprise Design Procedures for New Learning Systems

*Chapter 6 established the six-stage enterprise workflow, Plan, Design, Develop, Validate, Implement, Learn, that governs instructional work under IDOS. That workflow does not change depending on what is being built. What changes is the object: a full course is a different design problem than a single lesson, a learning pathway, a competency-based program, or a five-minute performance-support job aid.*

*This chapter applies the six-stage workflow to six design objects an organization commonly builds, and gives each one the one piece of guidance that is genuinely specific to it: a distinctive internal flow model. The procedural scaffolding, phase checkpoints, engine responsibilities, AI-assisted activity tables, is identical to Chapter 6 and is not repeated here; readers implementing any of the six objects should combine this chapter's flow model with Chapter 6's workflow and Chapter 2's engine domains.*

#### 8.1 Six Design Objects, One Workflow

Table 8.1 summarizes the six instructional design objects covered in this chapter, the primary specification each produces, and where each fits relative to the others. Complexity increases moving down the table: a course is typically composed of modules, modules of lessons, and a learning pathway or competency program sequences multiple courses toward a larger credential or capability.

| Design object | Primary output   | Typical scope                                 |
|---------------|------------------|-----------------------------------------------|
| Course        | Course Blueprint | A complete, standalone instructional offering |

|                                     |                                     |                                                         |
|-------------------------------------|-------------------------------------|---------------------------------------------------------|
| Module                              | Module Blueprint                    | A building block within an approved course              |
| Lesson                              | Lesson Blueprint                    | A single learning event within a module                 |
| Learning Pathway                    | Learning Path Blueprint             | A multi-course learner journey toward a destination     |
| Competency-Based Program            | Competency-Based Learning Blueprint | Progression governed by demonstrated mastery, not time  |
| Microlearning / Performance Support | Microlearning Blueprint             | A short, just-in-time intervention at the point of need |

*Table 8.1. The six enterprise design objects and their relationship.*

Each object still moves through Plan → Design → Develop → Validate → Implement → Learn (Chapter 6, Table 6.1), still engages all five operational engines (Chapter 2), and still produces the same five categories of enterprise output – specification, supporting package, decision log, traceability record, approved baseline (Table 2.2). What is genuinely distinct about each object is summarized in the sections below.

That engagement is concrete, not just asserted: each design object draws on a predictable subset of the twenty Core Agent Roles from Section 2.9, not all twenty at once. As a starting map, not a fixed rule, organizations should configure agents to the project in front of them, not to this list:

- Course and Module (Section 8.2) – the Instructional Planning Agent drafts architecture and sequence; the Content Development Agent produces the resulting materials once outcomes are approved.
- Lesson (Section 8.3) – the Content Development Agent drafts the Connect through Extend stages; the Assessment Design Agent supports the Assess stage specifically.

- Learning Pathway (Section 8.4) – the Curriculum Design Agent maps the multi-course structure; the Knowledge Intelligence Engine's Learning Analytics Agent (Section 2.9) supports the Progress stage once the pathway is live, the one point in Chapter 8 where a non-HDE, non-AIPE agent plays a defined role.
- Competency-Based Program (Section 8.5) – the Curriculum Design Agent maps required competencies; the Assessment Design Agent's involvement is sequenced deliberately early, consistent with this section's own “evidence before content” commitment.
- Microlearning and Performance Support (Section 8.6) – the Research & Analysis Agent supports the Identify stage's task and gap analysis; the Content Development Agent drafts the single Learn/Apply artifact that follows.

## 8.2 Course and Module Design

Course and module design apply the standard workflow with minimal variation. The one design decision that deserves emphasis: every module learning outcome should trace to one or more approved course learning outcomes before module design proceeds, and every lesson within a module should trace the same way to the module. Skipping this traceability check is the most common cause of incoherent courses, modules that are individually well designed but collectively redundant, gapped, or misaligned with what the course claims to teach.

- Course-level design establishes the overall architecture: learning outcomes, sequencing, assessment strategy, and the modules that will populate it.
- Module-level design inherits course outcomes and adds nothing that does not trace back to at least one of them.

Beyond this traceability discipline, course and module design do not require a distinct flow model – Chapter 6's six stages apply directly.

### 8.3 Lesson Design — The IDOS Lesson Flow

A lesson is where instructional design becomes a lived experience for a learner, and it is the one design object where a dedicated internal flow model earns its place. The IDOS Lesson Flow organizes a single learning event into eight stages, grounded in established learning-science principles, prior-knowledge activation (Merrill, 2002), cognitive load management (Mayer, 2021; Sweller et al., 1998), and retrieval-based assessment (Roediger & Karpicke, 2006); see also Section 2.3's Learning Science Architecture domain, which cites the same literature underlying the Human Design Engine's learning-science domain.

| Stage    | Purpose                                           | Learning-science focus                         |
|----------|---------------------------------------------------|------------------------------------------------|
| Connect  | Activate prior knowledge, establish relevance     | Prior-knowledge activation, advance organizers |
| Engage   | Capture attention, establish psychological safety | Motivation, emotional engagement               |
| Explore  | Introduce new concepts                            | Cognitive load management, worked examples     |
| Practice | Build confidence and competence with support      | Deliberate practice, scaffolding, feedback     |
| Apply    | Transfer knowledge to authentic situations        | Authentic learning, problem-solving            |
| Reflect  | Evaluate understanding and strategy               | Metacognition, self-regulated learning         |
| Assess   | Gather evidence of learning                       | Formative assessment, retrieval practice       |
| Extend   | Reinforce and prepare for what's next             | Spaced learning, transfer                      |

*Table 8.2. The IDOS Lesson Flow: eight stages, each with a distinct learning-science rationale.*

A lesson designer working through Chapter 6's Design stage should populate each of these eight stages deliberately rather than defaulting to a content dump followed by a quiz. The Connect and Reflect stages are often overlooked in instructional practice, despite research emphasizing the importance of activating prior knowledge, integrating new learning with existing knowledge, and engaging learners in reflective processing to strengthen understanding, transfer, and long-term retention (Denton, 2011; Merrill, 2002).

## 8.4 Learning Pathway Design — The IDOS Learner Journey

A learning pathway is not a longer course; it is a different kind of object entirely, because it must accommodate learners entering at different points, progressing at different rates, and exiting into different destinations (credentials, employment, further study). The IDOS Learner Journey organizes pathway design around eight developmental stages rather than instructional units.

| Stage       | Enterprise focus                                         |
|-------------|----------------------------------------------------------|
| Enter       | Eligibility, prior-learning recognition, learner goals   |
| Orient      | Expectations, pathway structure, advising                |
| Learn       | Structured courses, modules, and lessons                 |
| Practice    | Guided application, projects, workplace experience       |
| Demonstrate | Authentic assessment and portfolios                      |
| Progress    | Progress review, analytics, intervention                 |
| Credential  | Badges, certificates, academic credit                    |
| Advance     | Career progression, further education, alumni engagement |

*Table 8.3. The IDOS Learner Journey: eight developmental stages for pathway design.*

The design implication that most distinguishes pathway work from course work: a pathway blueprint must specify learner support (advising, mentoring, intervention triggers) with the same rigor as instructional content. A pathway with excellent courses but inadequate advising, onboarding, and progress support is more likely to experience learner attrition than one that integrates structured learner support throughout the pathway (Bailey et al., 2015). Learner support architecture is therefore not an add-on to pathway design but a co-equal design object alongside instructional content and assessment.

## 8.5 Competency-Based Program Design – The IDOS Competency Mastery Framework

Competency-based design inverts the usual instructional design question. Instead of “what should learners study,” it begins with “what must learners be able to do, and how will that be verified with evidence.” The IDOS Competency Mastery Framework reflects that inversion.

| Stage       | Enterprise focus                                                         |
|-------------|--------------------------------------------------------------------------|
| Identify    | Define required competencies against workforce or professional standards |
| Develop     | Build knowledge and skill through structured learning                    |
| Practice    | Deliberate practice, simulation, workplace experience                    |
| Demonstrate | Authentic evidence of achieved competency                                |
| Validate    | Independent verification against quality standards                       |
| Credential  | Formal recognition of verified mastery                                   |
| Apply       | Transfer to authentic professional practice                              |
| Advance     | Progress to greater responsibility or specialization                     |

*Table 8.4. The IDOS Competency Mastery Framework: eight stages organized around verified performance rather than instructional time.*

Three design commitments follow directly from this inversion, and all three echo principles already established elsewhere in this book. Competencies must be observable and measurable, not just intended (the same discipline the Human Design Engine applies to learning outcomes generally, Section 2.3). Evidence and assessment design must be planned before instructional content is developed, not after, the reverse of how many organizations build training. And competency verification should be independent of the team that developed the instructional content, for exactly the reason the Quality Intelligence Engine is kept independent of the AI Instructional Production Engine in Chapter 2: the reviewer should not be grading their own work.

## 8.6 Microlearning and Performance Support – The IDOS Performance Support Framework

Microlearning inverts the usual design question a second time. Course, module, and lesson design ask “what should learners know.” Microlearning asks “what must someone do better, right now, at their desk or on the floor.” The IDOS Performance Support Framework reflects that shift from instruction toward point-of-need enablement.

| Stage     | Enterprise focus                                       |
|-----------|--------------------------------------------------------|
| Identify  | The specific task or performance gap requiring support |
| Access    | Rapid, intuitive retrieval at the moment of need       |
| Learn     | Only the essential information required for the task   |
| Apply     | Immediate use in authentic work                        |
| Reinforce | Spaced reminders and follow-up                         |
| Retrieve  | Retrieval practice for long-term retention             |
| Perform   | Accurate, confident task execution                     |
| Improve   | Performance data feeding future refinement             |

*Table 8.5. The IDOS Performance Support Framework: eight stages organized around task performance rather than content coverage.*

A useful diagnostic when a microlearning project starts to drift: if the resource under development is teaching a topic rather than solving a task, it has quietly become a miniature course, and the design team should confirm whether Chapter 6's full workflow, not this abbreviated model, is actually the right tool for the underlying need.

## 8.7 One Pattern, Four Names

A reader who has just worked through Sections 8.3–8.6 will notice that the Lesson Flow, Learner Journey, Competency Mastery Framework, and Performance Support Framework are all eight stages long, and all move through roughly the same shape: orient the learner, build capability, prove it, and advance. That similarity is not accidental, and this book is better served by naming it than by hoping no one notices.

| Shared stage function | Lesson Flow      | Learner Journey                 | Competency Mastery           | Performance Support  |
|-----------------------|------------------|---------------------------------|------------------------------|----------------------|
| Orient                | Connect          | Enter / Orient                  | Identify                     | Identify             |
| Build                 | Engage / Explore | Learn                           | Develop                      | Access / Learn       |
| Practice              | Practice         | Practice                        | Practice                     | Apply                |
| Prove                 | Apply / Reflect  | Demonstrate                     | Demonstrate / Validate       | Reinforce / Retrieve |
| Advance               | Assess / Extend  | Progress / Credential / Advance | Credential / Apply / Advance | Perform / Improve    |

*Table 8.6. The shared five-function skeleton underlying all four Chapter 8 flow models.*

Given that they share a skeleton, the honest question is why this book gives them four names instead of one. The answer is practitioner communication, not structural necessity: a lesson designer and a credentialing-program architect are solving recognizably different problems, with different stakes, different vocabularies, and different

stakeholders in the room, telling a compliance-training team to consult “the generic eight-stage model” produces less uptake than telling them to consult “the Lesson Flow.” The four names exist to be memorable and specific to the object being designed; Table 8.6 exists so that the underlying consistency is stated rather than merely implied. If a future edition of this framework identifies a fifth design object, it should get a fifth adapted model built on the same skeleton, not a fifth unrelated invention.

## 8.8 Chapter Summary

Six design objects, course, module, lesson, learning pathway, competency-based program, and microlearning, all move through the same six-stage enterprise workflow (Chapter 6) and the same five operational engines (Chapter 2). What distinguishes each object is captured in this chapter as a compact internal flow model: the Lesson Flow for single learning events, the Learner Journey for multi-course pathways, the Competency Mastery Framework for evidence-based programs, and the Performance Support Framework for point-of-need interventions – four names built on one shared skeleton (Table 8.6), not four unrelated inventions. Organizations implementing any of these six objects should pair the relevant flow model here with the governance, quality, and knowledge-management machinery already established in Part II, rather than treating each design object as requiring its own separate operating system.

## Chapter 9

### Enterprise Modernization Methodology

*Chapter 8 addressed designing new learning systems. Most instructional work, in practice, is not new design – it is deciding what to do with something that already exists. This chapter provides IDOS's methodology for modernizing existing learning resources: how to evaluate them, how much to change, which parts to keep, and how to govern the update with the same rigor as new design.*

*This chapter deliberately reuses rather than restates: readiness follows the five-pillar model from Chapter 4, risk follows the same discipline as Chapter 4's risk register, and governance follows Chapter 5. What is genuinely distinct about instructional modernization, and covered in full here, is the decision logic for how much to change and what to do with existing assets.*

#### 9.1 Guiding Principles of Instructional Modernization

Instructional modernization is governed by ten principles. Several directly extend commitments already established elsewhere in this book; the table below notes those connections rather than re-arguing them.

| Principle                    | Enterprise implication                                                                           |
|------------------------------|--------------------------------------------------------------------------------------------------|
| Evidence before change       | Modernization decisions rest on performance data and feedback, not preference                    |
| Preserve before replace      | Existing assets are kept whenever they still work (see Legacy Asset Classification, section 9.3) |
| Proportional modernization   | Effort matches the size of the problem (see Maturity Model, section 9.2)                         |
| Learner-centered improvement | Learner outcomes remain the primary success measure                                              |
| Strategic alignment          | Modernization supports organizational mission and external requirements                          |

|                                    |                                                       |
|------------------------------------|-------------------------------------------------------|
| Responsible Human–AI collaboration | Same authority model as Chapter 2 (Table 2.2)         |
| Accessibility by design            | Integrated throughout, not a final compliance pass    |
| Governance throughout              | Applied continuously, consistent with Chapter 5       |
| Continuous modernization           | An ongoing capability, not a one-time project         |
| Knowledge preservation             | Feeds the Knowledge Intelligence Engine (Section 2.7) |

*Table 9.1. Guiding principles of enterprise instructional modernization.*

## 9.2 The Instructional Modernization Maturity Model

Not every outdated learning resource needs to be rebuilt from scratch. Drawing on curriculum update, redesign, and educational change literature, the Instructional Modernization Maturity Model (IMMM) organizes modernization into four progressively more intensive levels of intervention, enabling organizations to match the scope of change to instructional and organizational need rather than defaulting to full redesign (Fowler et al., 2015; Hai-Jew, 2010, Thompson, 2017).

| Level        | Goal                                          | Typical scope                                                                |
|--------------|-----------------------------------------------|------------------------------------------------------------------------------|
| 1. Refresh   | Maintain relevance                            | Content corrections, updated references, formatting, media swaps             |
| 2. Enhance   | Improve quality without changing architecture | Accessibility, engagement, assessment, and AI-support improvements           |
| 3. Redesign  | Significantly improve effectiveness           | Reorganized content, revised activities and assessments, modernized strategy |
| 4. Transform | Re-engineer for new requirements              | Comprehensive redesign, technology migration, AI-enabled delivery            |

*Table 9.2. The four-level Instructional Modernization Maturity Model.*

Level should be selected using evidence from five dimensions, instructional impact, organizational impact, technology impact, resource investment, and implementation complexity, evaluated together rather than any single one in isolation. A resource that scores high on instructional impact but low on the other four dimensions is usually a Level 2 (Enhance) project wearing a Level 4 (Transform) request; separating these dimensions during scoping is what prevents the most common modernization failure mode, which is redesigning a resource far more extensively than its actual problems justify.

### 9.3 Legacy Asset Classification

Within any modernization project, individual assets, a learning outcome, an assessment, a case study, a video, deserve individual decisions rather than a single blanket verdict for the whole resource. Drawing on systematic instructional evaluation and curriculum revision principles, the Legacy Asset Classification Model provides five evidence-based modernization decisions for individual instructional assets (CAST, 2018, Dick et al., 2014, Mayer, 2021; Quality Matters, n.d.; Wiggins & McTighe, 2005; W3C, 2018).

| Decision | Meaning                                                                         |
|----------|---------------------------------------------------------------------------------|
| Preserve | Retain as-is; the asset remains effective and current                           |
| Enhance  | Targeted improvement while keeping the underlying design                        |
| Redesign | Substantial revision while preserving the instructional purpose                 |
| Retire   | Remove; the asset is obsolete or redundant                                      |
| Replace  | Substitute with a new asset; modernization cannot achieve the outcome otherwise |

*Table 9.3. The five Legacy Asset Classification decisions.*

Classification decisions should be based on evidence of instructional effectiveness, alignment with current learning outcomes, content

currency, learner value, accessibility, technology compatibility, maintainability, and organizational priorities rather than on prior investment or organizational attachment to existing assets. A polished case study built on outdated industry examples is not preserved out of respect for the effort that built it; it is enhanced, or it is retired, based on whether it still teaches what it needs to teach.

***Field Note (illustrative)***

*A hospital system's onboarding curriculum included a well-produced, expensive video module on electronic health record navigation — four years old, still visually polished, and quietly wrong about three menu paths after a system upgrade. The instinct was to Enhance it (new screen recordings, same structure); the evidence pointed to Retire, because the underlying workflow the video taught no longer existed. The video's production quality had nothing to do with whether it still taught the right thing, which is exactly the distinction Table 9.3 is built to force.*

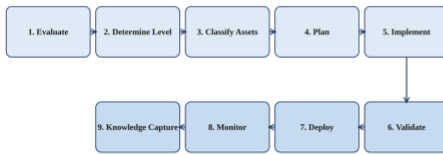
## 9.4 The Instructional Modernization Lifecycle

Instructional modernization follows a nine-stage lifecycle, distinct from Chapter 6's six-stage new-design workflow because it begins with an existing resource rather than a blank specification, and because two of its early stages, determining modernization level and classifying legacy assets, have no equivalent when nothing yet exists to evaluate.

| Stage                             | Purpose                                                       |
|-----------------------------------|---------------------------------------------------------------|
| 1. Evaluate                       | Establish an evidence-based baseline for the current resource |
| 2. Determine modernization level  | Apply the Maturity Model (section 9.2)                        |
| 3. Classify legacy assets         | Apply the Classification Model (section 9.3)                  |
| 4. Develop the modernization plan | Scope, priorities, resources, risk, success criteria          |
| 5. Implement improvements         | Execute the approved plan                                     |

|                                 |                                                                            |
|---------------------------------|----------------------------------------------------------------------------|
| 6. Validate                     | Independent confirmation of readiness<br>(consistent with Chapter 2's QIE) |
| 7. Deploy                       | Release into operational use                                               |
| 8. Monitor                      | Evaluate post-deployment performance                                       |
| 9. Enterprise knowledge capture | Feed lessons learned back into the organization<br>(Section 2.7)           |

*Table 9.4. The nine-stage enterprise instructional modernization lifecycle.*



Stages 6–9 map onto Chapter 6's Validate, Implement, and Learn stages (Table 6.1)

*Figure 9.1. The nine-stage lifecycle. Stages 6–9 (bottom row) map onto Chapter 6's Validate, Implement, and Learn stages (Table 6.1) — the same governance discipline applied to a resource that already exists.*

Stages 6 through 9 map directly onto Chapter 6's Validate, Implement, and Learn stages and should be governed in the same way. Independent quality review, formal quality checkpoints before release, and organizational knowledge capture are well-established principles of quality management and organizational learning (Argyris & Schön, 1978; Deming, 1986; Juran & Godfrey, 1999). Responsibility across all nine stages is distributed across the same five operational engines as every other IDOS activity, with the Human Design Engine leading strategic decisions, the Quality Intelligence Engine leading validation, and the Knowledge Intelligence Engine leading final capture.

As in Chapter 8, that engine involvement is carried out by specific Core Agent Roles (Section 2.9), not by the engines in the abstract. Evaluating a legacy asset (Stage 1) typically draws on the Human Design Engine's Research & Analysis Agent; revising content classified

for Enhance or Redesign (Stage 5) falls to the AI Instructional Production Engine's Content Development Agent, the same role Chapters 10 and 11 configure for new design; and enterprise knowledge capture (Stage 9) is the Knowledge Intelligence Engine's Knowledge Capture Agent, converting what Stage 1 through Stage 8 learned into the reusable asset Section 2.7 describes. The Quality Assurance Agent and Documentation Agent (Sections 2.9's QIE and AGE rows) support Stage 6's validation and the governance record required throughout, respectively.

## 9.5 Instructional Modernization Risk

Instructional modernization risk should be tracked using the same risk-register discipline established in Chapter 4 (Table 4.5), extended with categories specific to updating an existing resource rather than building a new one.

| Modernization-specific risk category     | Typical example                                                                                                |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Legacy asset misclassification           | A still-effective asset is discarded, or an obsolete one is preserved, based on sentiment rather than evidence |
| Scope creep toward Transform             | A Refresh-level need escalates into an unnecessary full redesign                                               |
| Instructional drift during modernization | Incremental changes accumulate until the resource no longer matches its original learning outcomes             |
| Legacy technology dependency             | The modernized resource still depends on infrastructure being retired elsewhere in the organization            |

*Table 9.5. Risk categories specific to modernization, additive to Chapter 4's general implementation risk table.*

## 9.6 Instructional Modernization Quality Assurance

Quality assurance for instructional modernization follows the same principle established for the Quality Intelligence Engine in Chapter 2: independent validation, evidence-based findings, and a formal implementation readiness decision rather than a routine approval (Deming, 1986; Juran & Godfrey, 1999; Quality Matters, n.d.). The one dimension worth adding specifically for modernization is consistency, verifying that a partially modernized resource does not leave learners moving between an updated section and an untouched one with jarring differences in terminology, design, or expectations. A modernization project that improves 70% of a course while leaving the remaining 30% untouched should be validated as a whole experience, not as two separate projects that happen to share a course shell.

## 9.7 AI Governance for Modernization

AI-supported instructional modernization, content revision, accessibility improvement, consistency checking, is governed by the same AI Governance Engine boundaries established in Chapter 2, formalized at the organizational level through the governance policy domains introduced in Section 5.4. No modernization-specific governance framework is needed beyond applying those existing policies to a new context; the risk of drafting a second, parallel AI policy specifically for modernization work is that the two policy sets drift apart over time, which itself becomes exactly the kind of inconsistency risk this chapter warns against in Section 9.6 (ISO, 2023).

## 9.8 Chapter Summary

Modernization is a distinct nine-stage lifecycle built around two decisions that new design does not require: how much to change (the four-level Maturity Model) and what to do with each existing asset (the five-way Legacy Asset Classification). Readiness, risk, governance, and

quality assurance all extend the models already established in Chapters 4, 5, and 2 rather than introducing parallel frameworks, modernization is best understood as IDOS applied to a resource that already exists, not as a separate operating system requiring its own governance structure.

PART IV

## Applied Case Studies

## Chapter 10

### IDOS Applied: Redesigning a Compliance Training Module

*Parts I through III described IDOS in the abstract. This chapter and Chapter 11 apply it to two examples of different kinds, this one a composite synthesized from well-documented, common patterns in AI-assisted compliance training, Chapter 11 a composite synthesized from well-documented patterns in higher education, so the framework can be evaluated against something concrete, in more than one sector, rather than only as a description of itself.*

**About This Case Study:** This chapter presents an illustrative case study developed from well-established patterns in AI-assisted instructional design, compliance training, and AI governance rather than from a single organizational implementation. The scenario is intended to demonstrate how the IDOS framework operates in practice while avoiding the disclosure of proprietary organizational information. The AI-generated regulatory error and the governance response presented in the case illustrate documented risks associated with using generative AI for high-stakes factual content without appropriate human oversight and governance (ISO, 2023; NIST, 2023; OECD, 2019). No organizational outcome data (e.g., completion rates, performance improvements, error reduction, or cost savings) are presented or implied. The purpose of this chapter is not to evaluate the effectiveness of IDOS empirically but to demonstrate its architecture through a realistic instructional modernization scenario. Section 10.9 returns to this distinction.

#### 10.1 Case Overview

A mid-size financial services firm needs to refresh its annual Anti-Money Laundering (AML) awareness training for frontline staff. The

existing course is a 45-minute narrated slide deck, last substantively revised four years ago, with a stale multiple-choice quiz that most staff pass on a second attempt without demonstrating real understanding. The compliance team wants to shorten the course, replace passive narration with short scenario-based decisions, and use AI to accelerate drafting, but the compliance officer is, appropriately, wary of AI generating content involving actual regulatory thresholds and reporting obligations without a governed review process. This is precisely the situation IDOS is designed for: real productivity pressure, real AI capability, and real accountability risk, together.

In Chapter 8's terms (Table 8.1), this is a Module-level redesign: a self-contained, single-sitting training unit rather than a full multi-week course, a standalone lesson, or a pathway. It uses the standard six-stage workflow (Chapter 6) rather than the Lesson Flow specifically, because the redesign spans the whole unit, outcomes, assessment, and delivery together, rather than a single learning event within a larger structure. Section 10.7 adds a second, shorter example built on a different design object, so this chapter demonstrates the framework flexing across more than one case.

## 10.2 Stage 1: Human Design Engine

Before any AI drafting begins, the instructional designer and the compliance officer complete the Human Design Specification together. Table 10.2 covers all ten domains from Section 2.3 — but not every domain carries an equally distinctive decision for this case. Where a domain reflects a genuine, case-specific choice, the entry says so in full; where it reflects standard practice this project shares with almost any comparable one, the entry is brief and marked accordingly, rather than manufacturing false specificity to look more substantial than it is.

| Domain (Section 2.3)                             | AML Awareness Refresh                                                                                                                                                                              |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Performance Need Architecture                 | Staff pass the current quiz without reliably recognizing suspicious-activity indicators in realistic scenarios; over-reliance on definitional recall.                                              |
| 2. Learner and Stakeholder Architecture          | Frontline branch and call-center staff; high turnover; prior exposure to one earlier AML course; mixed confidence with financial terminology; compliance officer as subject-matter authority.      |
| 3. Performance Outcome Architecture              | Given a realistic transaction scenario, staff correctly identify escalation criteria and cite the specific triggering indicator.                                                                   |
| 4. Assessment Architecture                       | Scenario-based decision items; passing requires correct escalation decisions on novel scenarios, not memorized thresholds; immediate feedback explains the correct indicator.                      |
| 5. Learning Science Architecture (routine)       | Standard retrieval-practice and worked-example sequencing; not a distinctive pedagogical choice for this project.                                                                                  |
| 6. Instructional Strategy Architecture (routine) | Scenario, decision point, feedback — a standard sequence, not a distinctive choice here.                                                                                                           |
| 7. Content Architecture                          | Approved sources limited to internal compliance policy and current regulatory guidance supplied by the compliance officer; excluded: general AML theory not tied to a specific frontline decision. |
| 8. Learning Experience Architecture (routine)    | Follows directly from the instructional strategy above; no separate decision required.                                                                                                             |
| 9. Media Strategy Architecture                   | Text-based scenario cards, no video, chosen specifically to keep the module usable on mobile devices during shift breaks.                                                                          |
| 10. Success Architecture (routine)               | Standard compliance-officer sign-off; success measured by post-module assessment accuracy.                                                                                                         |

*Table 10.2. Human Design Specification, all ten domains — AML Awareness Refresh. Domains marked “routine” reflect standard practice rather than a case-specific decision.*

### Human Design Specification — a closer look at the content boundary

**Content boundary (Domain 7, elaborated):** All regulatory thresholds and reporting timeframes must match current internal compliance policy exactly; AI-drafted scenarios may not introduce or imply a threshold not supplied by the compliance officer.

That line, the content boundary, is the single most consequential decision in this specification, and it is made here, before any AI drafting occurs, precisely because it is a compliance decision, not a production one. Section 10.4 shows why.

### 10.3 Stage 2: AI Governance Engine

The Human Design Specification's content boundary becomes an enforceable rule in the AI Governance Specification, which also defines what the AI Instructional Production Engine may and may not do unsupervised. Table 10.3 covers all ten governance domains from Section 2.4; as in Table 10.2, routine domains are marked and kept brief rather than padded.

| Domain (Section 2.4)                           | AML Awareness Refresh                                                                                                                                                                           |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Governance Objective Architecture (routine) | Accurate AI-assisted drafting without unsupported claims — the standard objective for this class of project.                                                                                    |
| 2. Governance Authority Architecture (routine) | Compliance officer over regulatory content; designer over instructional structure — a standard split, not unique here.                                                                          |
| 3. AI Operating Boundary Architecture          | AI may draft scenario narrative, distractors, and plain-language explanations from supplied source material; AI may not generate or infer specific regulatory figures, deadlines, or citations. |
| 4. Risk and Compliance Architecture (routine)  | Primary risk is regulatory misstatement reaching frontline staff as authoritative guidance — the standard risk for AI-assisted compliance content.                                              |
| 5. Data and Privacy Architecture               | No customer or transaction data used in training materials; all scenarios built from de-identified, composite situations.                                                                       |

|                                                          |                                                                                                                      |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 6. Transparency and Documentation Architecture (routine) | Standard logging of source-material reference and reviewer sign-off before publication.                              |
| 7. Human Oversight Architecture                          | 100% of AI-drafted scenario content reviewed by the compliance officer before proceeding to quality validation.      |
| 8. Quality Governance Architecture (routine)             | Governance sign-off precedes quality review, as in every IDOS project (Section 2.8).                                 |
| 9. Change Management Architecture                        | Any AI-drafted content implying an unsupplied threshold is logged as a governance exception, not silently corrected. |
| 10. Governance Improvement Architecture (routine)        | Quarterly review of exceptions; standard cadence, not a distinctive decision here.                                   |

*Table 10.3. AI Governance Specification, all ten domains — AML Awareness Refresh. Domains marked “routine” reflect standard practice rather than a case-specific decision.*

This is the practical meaning of “governance precedes production” from Section 3.3: the rule about thresholds (Domain 3) exists in an approved document before a single scenario is drafted, rather than being applied ad hoc if someone happens to notice a problem during review.

## 10.4 Stage 3: AI Instructional Production Engine – A Governed Draft in Practice

Table 10.4 covers all nine production domains from Section 2.5. Section 2.5 itself notes that these domains “function more like workflow stages than independent decision areas” — and that shows here: most of this table is routine execution, with one domain (Asset refinement) carrying the entire chapter's central finding, shown in detail afterward. A single agent role, the Content Development Agent (Section 2.9), was configured and restricted to the approved source list for this project; no other agent role was required for a module this compact.

| Domain (Section 2.5)                                     | AML Awareness Refresh                                                                                                                                                                                                                            |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Production Planning (routine)                         | Six scenarios, four-week window — scope and schedule, not a distinctive decision.                                                                                                                                                                |
| 2. Production Workflow Orchestration (routine)           | Batches of two, reviewed before the next batch — standard staged production.                                                                                                                                                                     |
| 3. Instructional Content Production (routine)            | Scenario cards and feedback text drafted by the Content Development Agent, revised by the designer — standard production flow.                                                                                                                   |
| 4. Assessment and Learning Resource Production (routine) | Assessment items generated from the same scenario bank by the same agent; standard alignment check against the blueprint.                                                                                                                        |
| 5. Media and Multimedia Production (routine)             | Scenario cards assembled as text-based media, consistent with the mobile-first delivery decision in Section 10.2, Domain 9; no video or narrated media required for a module this compact.                                                       |
| 6. Asset Refinement                                      | The compliance officer's threshold correction, shown in detail below, is the representative example of this domain, and of this chapter, in action — subject-matter refinement of an AI draft before it moves to independent quality validation. |
| 7. Production Documentation (routine)                    | Standard logging of source, date, and reviewer for every scenario.                                                                                                                                                                               |
| 8. Asset Integration (routine)                           | Standard assembly and sequence testing within the compliance platform.                                                                                                                                                                           |
| 9. Production Readiness (routine)                        | Standard readiness confirmation before handoff to quality review, on schedule in week four.                                                                                                                                                      |

*Table 10.4. AI Instructional Production Specification, all nine domains — AML Awareness Refresh. Domains marked “routine” reflect standard practice rather than a case-specific decision.*

#### **AI draft (before human review)**

Scenario: A customer attempts to deposit \$12,500 in cash and asks the teller not to ‘make a big deal of it.’ Because this exceeds the \$15,000 reporting threshold, staff should escalate immediately.

**Compliance officer's revision (after human review)**

Scenario: A customer attempts to deposit \$12,500 in cash and asks the teller not to 'make a big deal of it.' This exceeds the \$10,000 currency transaction reporting threshold under the Bank Secrecy Act, and the customer's request not to draw attention is itself a behavioral indicator independent of the dollar amount. Staff should escalate for both reasons.

**Reviewer note logged:** *AI draft stated an incorrect threshold (\$15,000 instead of \$10,000) and missed the behavioral indicator entirely. Corrected before release; logged per the governance exception trigger in Section 10.3.*

*A note on the \$10,000 figure itself: it reflects the U.S. Bank Secrecy Act's currency transaction reporting threshold at the time of writing. Regulatory thresholds change, and this book is not compliance guidance — an organization using this example as a template should verify current figures against the applicable regulation, not against this page, regardless of which number the AI or the human reviewer produced.*

This is not a hypothetical risk invented for this book, confidently stated, plausible-sounding, and wrong regulatory detail is one of the most-cited failure modes of ungoverned generative AI use in high-stakes content (Ji et al., 2023; NIST, 2023). The AI Governance Engine did not prevent the AI from making the error; nothing in current AI capability guarantees that. What it did was guarantee that a qualified human reviewed every scenario before release, and logged the exception rather than letting a quiet correction disappear without a record, which is what makes it available to the Knowledge Intelligence Engine in Section 10.6.

## 10.5 Stage 4: Quality Intelligence Engine

Independent quality review, conducted by a reviewer who was not involved in drafting, evaluates the revised course against the Human Design Specification's outcomes, accessibility requirements, and governance compliance. Table 10.5 covers all ten quality domains from Section 2.6; unlike Table 10.4, most of these domains carry a genuine

finding rather than routine confirmation, because validation is where this project's specific issues surfaced.

| Domain (Section 2.6)                         | AML Awareness Refresh                                                                                                                                                                                                             |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Validation Planning (routine)             | Standard scope — instructional alignment, governance compliance, accessibility; reviewer independent of drafting.                                                                                                                 |
| 2. Instructional Alignment Validation        | Pass. Scenario items assess applied escalation judgment, consistent with the performance outcome in Section 10.2, not definitional recall.                                                                                        |
| 3. Governance Compliance Validation          | Pass. One governance exception on record (Section 10.4), correctly logged and resolved prior to this review.                                                                                                                      |
| 4. Production Integrity Validation (routine) | Pass, by standard record check — every scenario traceable to its source reference and reviewer.                                                                                                                                   |
| 5. Accessibility and Inclusion Validation    | Conditional. The Accessibility Agent (Section 2.9) drafted alt text and screen-reader-tested navigation for every scenario card; the independent quality lead's certification found two decorative images still lacking alt text. |
| 6. Quality Evidence and Metrics (routine)    | Standard evidence sources: production record, governance decision log, screen-reader walkthrough.                                                                                                                                 |
| 7. Risk and Issue Validation                 | One open issue (missing alt text), low severity, does not block a conditional release.                                                                                                                                            |
| 8. Corrective Actions and Recommendations    | Add alt text to the two flagged images before publication; no other changes required.                                                                                                                                             |
| 9. Implementation Readiness Validation       | Conditional pass — release approved pending the alt-text correction, verified by the quality lead without a full re-review.                                                                                                       |
| 10. Continuous Quality Improvement           | Recommend adding an automated alt-text check to the production checklist so this finding does not recur.                                                                                                                          |

*Table 10.5. Quality Intelligence Specification, all ten domains — AML Awareness Refresh. Domains marked “routine” reflect standard practice rather than a case-specific finding.*

A conditional pass with one specific corrective action (Domain 8) is a more realistic and more useful outcome than a blanket approval, and it is the kind of finding an independent reviewer is positioned to catch

precisely because they were not the one focused on drafting scenario content. Chapter 11's quality review surfaces the same category of defect, an accessibility, not instructional-content, issue, and that repetition is not manufactured for this book: heading-structure and navigation defects are among the most commonly reported issues in real learning-management-system content, in any sector (WebAIM, 2024).

## 10.6 Stage 5: Knowledge Intelligence Engine

The governance exception logged in Section 10.4 becomes a reusable organizational asset, not just a resolved incident. Table 10.6 covers all ten knowledge domains from Section 2.7; as in Table 10.5, most entries here carry real content because this stage exists specifically to capture what was distinctive about the project.

| Domain (Section 2.7)                  | AML Awareness Refresh                                                                                                                                      |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Knowledge Acquisition (routine)    | Standard capture of the governance exception and accessibility finding at project close.                                                                   |
| 2. Knowledge Organization (routine)   | Standard filing under existing repository categories.                                                                                                      |
| 3. Knowledge Validation (routine)     | Standard reviewer sign-off before repository publication.                                                                                                  |
| 4. Knowledge Asset Management         | The revised scenario, corrected threshold, added behavioral indicator, retained as a reusable example for future scenario drafting.                        |
| 5. Pattern and Trend Analysis         | This is the compliance team's second logged instance of AI-drafted content misstating a specific figure, suggesting a recurring rather than isolated risk. |
| 6. Best Practices and Lessons Learned | Numeric regulatory figures should be supplied as literal source text, never generated or paraphrased by AI, even when the AI's answer is likely correct.   |
| 7. Decision Intelligence              | This finding informs the decision to require literal source figures in every future AI-assisted regulatory drafting project, not only this one.            |

|                                                        |                                                                                                                                                                                             |
|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8. Innovation and Improvement                          | Proposed pilot: an automated pre-review check that flags any AI-drafted numeric claim not traceable to the approved source list, before human review begins.                                |
| 9. Knowledge Sharing and Reuse                         | Rule applied to the AI Governance Specification for the following year's refresh before drafting begins, rather than rediscovered during review.                                            |
| 10. Organizational Learning and Capability Development | Across two production cycles, the compliance team's AI-assisted drafting has shifted from catching regulatory errors during review to preventing them through an upstream content boundary. |

*Table 10.6. Knowledge Intelligence Specification, all ten domains – AML Awareness Refresh. Domains marked “routine” reflect standard practice rather than a case-specific entry.*

This is the organizational-learning claim from Section 2.7 made concrete: the lesson does not live only in one reviewer's memory. It becomes a standing governance rule the next project inherits automatically, which is the specific mechanism by which IDOS claims to make an organization's AI-assisted instructional work safer over time rather than only as safe as its most careful reviewer on a given day.

## 10.7 A Second Design Object: The Follow-Up Job Aid

The module redesign in Sections 10.1–10.6 demonstrates one design object from Chapter 8's list of six. A framework that claims to flex across course, module, lesson, pathway, competency program, and microlearning should show more than one, so this section adds a short, second example built on a genuinely different object: a microlearning performance-support job aid, designed using the Performance Support Framework (Section 8.6) rather than the six-stage module workflow used above.

Three months after the AML module launched, branch staff kept asking the same follow-up question during audits: what to do when a flagged transaction involves a long-standing customer the teller personally knows. Building a new course module for one recurring

question would be disproportionate — exactly the diagnostic Section 8.6 describes. The compliance team instead built a one-screen job aid.

#### **Microlearning Blueprint (excerpt) — Familiar-Customer Escalation Job Aid**

**Identify (Performance Support Framework):** Staff hesitate to escalate flagged transactions involving customers they recognize, worried escalation implies personal accusation.

**Access:** Delivered as a single searchable card inside the teller platform, surfaced automatically when a transaction is flagged, not as a separate course to locate.

**Learn / Apply:** One sentence: ‘Escalation evaluates the transaction, not the customer — recognizing someone is never a reason to skip a required step.’ One example, no quiz.

**Governance boundary reused:** The same AI Governance Specification rule from Section 10.3 applied unchanged: AI drafted the plain-language explanation; the compliance officer supplied and verified the one procedural claim it makes.

The point of including this second, much smaller example is not the job aid itself – it is that the same governance boundary, the same human-review requirement, and the same knowledge-capture habit from Sections 10.2–10.6 ported directly to a completely different design object without modification. That portability, not any single artifact, is the claim Chapter 8 makes about the framework generalizing across the six design objects, and it is worth seeing demonstrated rather than only asserted.

## **10.8 Before/After: Governed vs. Ad Hoc AI Use**

Table 10.1 contrasts this governed redesign with how the same project frequently unfolds without an operating system like IDOS — not because ad hoc AI use is reckless by intent, but because without an explicit boundary, decisions like the threshold question in Section 10.4 have no guaranteed checkpoint at which they are caught.

| Dimension           | Ad hoc AI-assisted development                           | Under IDOS                                                           |
|---------------------|----------------------------------------------------------|----------------------------------------------------------------------|
| Regulatory accuracy | Depends on whichever reviewer happens to notice an error | Explicit governance rule; human review required by policy, not habit |
| Accountability      | Unclear who approved the final content                   | Traceable decision log from design through release                   |
| Reusable learning   | The lesson lives with whoever caught the error           | Logged as a governance rule inherited by future projects             |
| Quality check       | Often informal, by whoever is available                  | Independent review, separate from the drafting team                  |

*Table 10.1. Illustrative comparison of governed versus ad hoc AI-assisted development for this case.*

This comparison is illustrative, not a controlled study — no claim is made about how often ad hoc processes actually fail this way in practice, only that IDOS's specific design closes a gap that has no guaranteed safeguard without it.

## 10.9 What This Case Does, and Does Not, Show

This chapter demonstrates that IDOS's artifacts and checkpoints are concrete and usable: a Human Design Specification excerpt can be written in an afternoon by an instructional designer and a subject-matter expert together; a governance rule can be specific enough to catch a real category of error; a quality review can produce an actionable, partial finding instead of a binary pass or fail; and a lesson learned can be converted into a rule the next project inherits automatically.

It does not demonstrate that organizations using IDOS produce measurably better training, complete projects faster, or catch more errors than they would under a well-run ad hoc process with a diligent reviewer. No such comparative data exists yet for this framework.

Readers evaluating IDOS for adoption should treat this chapter as evidence of internal coherence, the pieces fit together and produce sensible artifacts, and should treat the Limitations and Future Research section in the back matter as the more important word on what remains unproven.

## 10.10 Chapter Summary

Run end to end, IDOS's five engines produced, for this illustrative case: a Human Design Specification that made a compliance-critical content boundary explicit before drafting began; an AI Governance Specification that turned that boundary into an enforceable, escalation-triggering rule; a production stage in which a real category of AI error was caught by required human review, carried out by the Content Development Agent (Section 2.9) within its governance boundary, rather than by chance; a quality review that produced a specific, actionable finding; and a knowledge-repository entry that converts one project's lesson into the next project's standing rule. Section 10.7 then showed the same governance boundary porting unmodified to a second, much smaller design object, evidence that the framework's claim to generalize across Chapter 8's six design objects is at least demonstrated twice in this book, not only once. The chapter's purpose was to make the framework's abstractions legible as documents a real team could produce, not to claim outcomes the framework has not yet been tested to deliver. Chapter 11 adds a third design object in a different sector, using the same composite technique: a higher-education AI practitioner certification program, reconstructed through the same five engines.

## Chapter 11

### IDOS Applied: Certifying AI Practitioners in Teaching

*Chapter 10 walked through a corporate compliance training redesign. This chapter applies the same five engines to a second case, in a different sector and around a different design object, so the framework can be evaluated against more than one kind of problem, still without exposing any single organization's specifics.*

**About This Case Study:** Like Chapter 10, this chapter presents a composite case study synthesized from common organizational approaches to faculty AI literacy, professional development, and institutional AI governance rather than from a single university implementation. The scenario is intended to demonstrate how the IDOS framework can be applied in practice while avoiding the disclosure of proprietary institutional information. The AI-generated compliance error and the governance response illustrate documented risks associated with using generative AI to develop high-stakes policy and compliance content without appropriate human oversight and governance, the same category of risk demonstrated in Chapter 10 through a different instructional context (NIST, 2023; OECD, 2019; ISO, 2023). No organizational outcome data (e.g., certification completion rates, policy compliance improvements, adoption rates, or cost savings) are presented or implied. The purpose of this chapter is not to evaluate the effectiveness of IDOS empirically but to demonstrate its architecture through a realistic faculty AI literacy and certification scenario. Section 11.8 returns to this distinction.

#### 11.1 Case Overview

A large public research university's center for teaching and learning found that AI adoption among faculty and instructional staff was

inconsistent and largely unguided. Some instructors had already integrated generative AI extensively into course design and grading workflows; others avoided it entirely out of uncertainty; and there was no shared, university-wide understanding of what responsible AI use in teaching actually required. The most consequential gap involved data privacy: faculty were pasting student work, essays, discussion posts, code, sometimes with names or student IDs still attached, into public AI tools to generate feedback, without a consistent understanding that this can violate the Family Educational Rights and Privacy Act (FERPA), the federal law governing the privacy of student education records.

In response, the center for teaching and learning replaced ad hoc departmental AI workshops with a single, university-wide AI Practitioner Certification: a competency-based badge available to faculty, instructional designers, and teaching staff, covering five competencies every certified practitioner must demonstrate — foundational AI literacy, AI-assisted instructional design, data privacy and compliance, academic integrity and transparency, and equity, accessibility, and critical evaluation. AI was used to accelerate the certification's own content development, under governance with real stakes: not a wrong number, as in Chapter 10, but a federal data-privacy requirement that AI drafting can just as easily soften into general encouragement.

In Chapter 8's terms (Table 8.1), this is a Competency-Based Program: progression governed by demonstrated mastery of the five competencies, not by seat time, using the Competency Mastery Framework (Section 8.5) rather than the module-level six-stage workflow used in Chapter 10's compliance case. Together, Chapter 10 and this chapter now demonstrate three of Chapter 8's six design objects, Module, Microlearning, and Competency-Based Program, across two sectors.

## 11.2 Stage 1: Human Design Engine

Before any AI drafting began, the instructional designer and the university's data governance office completed the Human Design Specification together. Table 11.2 covers all ten domains from Section 2.3; as in Chapter 10, routine domains are marked and kept brief rather than padded.

| Domain (Section 2.3)                             | AI Practitioner Certification Program                                                                                                                                                                                                                                            |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Performance Need Architecture                 | AI adoption among faculty and instructional staff was inconsistent and unguided, with no shared understanding of responsible practice — most consequentially, no consistent awareness that pasting identifiable student work into public AI tools can violate FERPA.             |
| 2. Learner and Stakeholder Architecture          | Faculty, instructional designers, and teaching staff seeking certification; the center for teaching and learning as program owner; the university's data governance office as subject-matter authority on FERPA and approved-tool requirements.                                  |
| 3. Performance Outcome Architecture              | Given a realistic teaching or course-design scenario, certified practitioners correctly apply the relevant competency, AI literacy, instructional design, data privacy, academic integrity, or equity and accessibility, rather than completing a reading with no applied check. |
| 4. Assessment Architecture                       | Scenario-based decision items across all five competencies, including one required case-response demonstrating correct data-handling for a common AI-assisted feedback scenario; certification tied to passing performance, not seat time.                                       |
| 5. Learning Science Architecture (routine)       | Worked examples and scenario practice; not a distinctive pedagogical choice for this program.                                                                                                                                                                                    |
| 6. Instructional Strategy Architecture (routine) | Five self-paced modules, one per competency, with a scenario checkpoint at the end of each — a standard structure.                                                                                                                                                               |
| 7. Content Architecture                          | Data-privacy and compliance content sourced only from the data governance office's current approved-AI-tool list and FERPA guidance; AI-drafted scenarios may not imply that a specific AI tool is safe for student data unless it appears on that list.                         |

|                                               |                                                                                                                                                                                                                 |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8. Learning Experience Architecture (routine) | Standard sequence: brief framing, scenario, decision point, feedback, and a closing summary of where to find the full approved-tool list.                                                                       |
| 9. Media Strategy Architecture (routine)      | Text-based scenario cards, no video, to keep the module usable across departments with varying technology access.                                                                                               |
| 10. Success Architecture                      | Certification requires sign-off from both the center for teaching and learning and the data governance office — a two-stakeholder approval this program specifically requires, unlike a single-approver module. |

*Table 11.2. Human Design Specification, all ten domains — AI Practitioner Certification Program. Domains marked “routine” reflect standard practice rather than a case-specific decision.*

**Human Design Specification — a closer look at the content boundary**

**Content boundary (Domain 7, elaborated):** Data-privacy and compliance content must match the university's current approved-AI-tool list and FERPA guidance exactly. AI-drafted scenarios may not imply that any AI tool is appropriate for identifiable student data unless that tool appears on the approved list supplied by the data governance office.

That boundary is the single most consequential decision in this specification, made here, before any AI drafting occurs, for the same reason Chapter 10's threshold boundary was: it is a legal-compliance decision, not a production one. Section 11.4 shows why.

**11.3 Stage 2: AI Governance Engine**

The Human Design Specification's content boundary becomes an enforceable rule in the AI Governance Specification. Table 11.3 covers all ten governance domains from Section 2.4.

|                      |                                       |
|----------------------|---------------------------------------|
| Domain (Section 2.4) | AI Practitioner Certification Program |
|----------------------|---------------------------------------|

|                                                          |                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Governance Objective Architecture (routine)           | Accurate, university-wide AI practitioner content without introducing compliance inaccuracies — the standard objective for this class of project.                                                                                                                                                                        |
| 2. Governance Authority Architecture                     | Data governance office holds final authority over all data-privacy and compliance content; instructional designer owns structure, pacing, and the other four competencies.                                                                                                                                               |
| 3. AI Operating Boundary Architecture                    | AI may draft scenario narrative, distractors, and plain-language explanations for the four non-compliance competencies from approved source material; for data-privacy content, AI may not generate, infer, or imply which AI tools are FERPA-appropriate — that list must come from the data governance office.         |
| 4. Risk and Compliance Architecture                      | Primary risk is not a wrong number, as in Chapter 10, but AI-drafted guidance that describes AI-assisted feedback generation as broadly helpful without the required qualification that only approved tools may be used with identifiable student work — a risk with real regulatory exposure, not only program quality. |
| 5. Data and Privacy Architecture (routine)               | No real student records used in training scenarios; all scenarios built from composite, de-identified situations.                                                                                                                                                                                                        |
| 6. Transparency and Documentation Architecture (routine) | Standard logging of source-material reference and reviewer sign-off before publication.                                                                                                                                                                                                                                  |
| 7. Human Oversight Architecture                          | 100% of AI-drafted data-privacy and compliance content reviewed by the data governance office before proceeding to quality validation; the other four competencies follow the standard single-reviewer path.                                                                                                             |
| 8. Quality Governance Architecture (routine)             | Governance sign-off is a precondition for the Quality Intelligence Engine's review, not a substitute for it.                                                                                                                                                                                                             |
| 9. Change Management Architecture                        | Any AI-drafted content endorsing AI-assisted work with student data without the approved-tool qualification is logged as a governance exception, not silently corrected.                                                                                                                                                 |
| 10. Governance Improvement Architecture (routine)        | Governance exceptions reviewed each term with the data governance office to determine whether the approved-tool list or its presentation needs to be more explicit.                                                                                                                                                      |

*Table 11.3. AI Governance Specification, all ten domains — AI Practitioner Certification Program. Domains marked “routine” reflect standard practice rather than a case-specific decision.*

This is “governance precedes production” (Section 3.3) again, in a third register: the rule about approved tools (Domain 3) exists in an approved document before a single scenario is drafted, exactly as the threshold rule did in Chapter 10 – the specific risk changes with the sector; the mechanism that catches it does not.

### 11.4 Stage 3: AI Instructional Production Engine – A Governed Draft in Practice

Table 11.4 covers all nine production domains from Section 2.5. As in Chapter 10, most of this table is routine execution, Section 2.5 describes these domains as workflow stages, not decision areas, with one domain carrying the chapter's central finding, shown in detail afterward. Two agent roles, named formally in Section 2.9, were configured for this program: the Content Development Agent drafted scenario narrative and explanations across all five modules, and the Assessment Design Agent drafted the scenario decision items.

| Domain (Section 2.5)                                     | AI Practitioner Certification Program                                                                                                                |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Production Planning (routine)                         | Five modules, one per competency, four-week production window.                                                                                       |
| 2. Production Workflow Orchestration (routine)           | Modules drafted in sequence, each reviewed before the next begins.                                                                                   |
| 3. Instructional Content Production (routine)            | Scenario cards, decision prompts, and feedback text drafted by the Content Development Agent and revised by the instructional designer.              |
| 4. Assessment and Learning Resource Production (routine) | Scenario-based assessment items generated by the Assessment Design Agent from the same source material, checked against the assessment blueprint.    |
| 5. Media and Multimedia Production (routine)             | Text-based scenario cards assembled for all five modules; no video or narrated media required for this program.                                      |
| 6. Asset Refinement                                      | The data governance office's correction to the AI-assisted feedback scenario, shown below, is the representative example of this domain, and of this |

|                                       |                                                                                                                  |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------|
|                                       | chapter, in action — subject-matter refinement of an AI draft before it moves to independent quality validation. |
| 7. Production Documentation (routine) | Standard logging of source, date, and reviewer for every scenario.                                               |
| 8. Asset Integration (routine)        | Modules assembled into the university's certification platform and sequence-tested before validation.            |
| 9. Production Readiness (routine)     | All five modules reviewed, logged, and assembled on schedule ahead of the certification's first cohort.          |

*Table 11.4. AI Instructional Production Specification, all nine domains — AI Practitioner Certification Program. Domains marked “routine” reflect standard practice rather than a case-specific decision.*

### **AI draft (before human review)**

Scenario: A colleague asks whether they can use a free AI writing tool to give quick feedback on student essays before grading. Explain that AI tools can be a great time-saver for generating feedback ideas and encourage them to try it.

### **Data governance office's revision (after human review)**

Scenario: A colleague asks whether they can use a free AI writing tool to give quick feedback on student essays before grading. AI tools can support feedback generation, but only tools on the university's current approved list may be used with any student work that includes a name, student ID, or other identifying information — uploading identifiable student work to a non-approved public AI tool creates FERPA exposure regardless of intent. De-identified work, or work already made public by the student, can be used more flexibly; when in doubt, check the approved-tool list or ask the data governance office before uploading anything.

**Reviewer note logged:** *AI draft endorsed AI-assisted feedback generally without the required data-handling qualification, which could lead a well-meaning colleague to unknowingly create FERPA exposure. Corrected before release; logged per the governance exception trigger in Section 11.3.*

*A note on the compliance framework described here: FERPA's requirements, a given university's approved-tool list, and its specific data governance procedures all vary and change over time. This book is not compliance or legal guidance — an*

*organization using this example as a template should verify current requirements against its own data governance office and legal counsel, not against this page.*

This is the same failure pattern Chapter 10 demonstrates, wearing a third face: confidently stated, plausible-sounding, and legally incomplete guidance is one of the most-cited risks of ungoverned generative AI use in compliance-adjacent content (Ji et al., 2023), and it is especially easy to miss here because the softened version reads as encouraging and helpful, not wrong, exactly the kind of omission a well-meaning reviewer might not catch without a content boundary stated in advance. The AI Governance Engine did not prevent the AI from generating it; nothing in current AI capability guarantees that. What it did was guarantee that a data governance-qualified reviewer saw every compliance-related scenario before release, and logged the exception rather than letting a quiet correction disappear, which is what makes it available to the Knowledge Intelligence Engine in Section 11.6.

## 11.5 Stage 4: Quality Intelligence Engine

Independent quality review, conducted by a reviewer not involved in drafting, evaluated the certification program against the Human Design Specification's outcomes, accessibility requirements, and governance compliance. Table 11.5 covers all ten quality domains from Section 2.6.

| Domain (Section 2.6)                  | AI Practitioner Certification Program                                                                                                                                       |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Validation Planning (routine)      | Standard scope: instructional alignment, governance compliance, accessibility; reviewer independent of drafting.                                                            |
| 2. Instructional Alignment Validation | Pass. Scenario items across all five competencies assess applied decision-making, consistent with the performance outcomes in Section 11.2, not passive reading completion. |

|                                              |                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3. Governance Compliance Validation          | Pass. One governance exception on record (Section 11.4), correctly logged and resolved by the data governance office prior to this review.                                                                                                                                                                                   |
| 4. Production Integrity Validation (routine) | Pass. Production record complete; every scenario traceable to its source reference and reviewer.                                                                                                                                                                                                                             |
| 5. Accessibility and Inclusion Validation    | Conditional. The Accessibility Agent (Section 2.9) drafted alt text and screen-reader-tested navigation for every scenario card; the independent quality lead's certification found the certification-completion page still uses non-sequential heading levels — a common LMS defect category, not specific to this program. |
| 6. Quality Evidence and Metrics (routine)    | Standard evidence sources: production record, governance decision log, screen-reader walkthrough.                                                                                                                                                                                                                            |
| 7. Risk and Issue Validation                 | One open issue (completion-page heading structure), low severity, does not block a conditional release ahead of the first cohort.                                                                                                                                                                                            |
| 8. Corrective Actions and Recommendations    | Correct the completion page's heading structure before the next certification cycle; no other changes required.                                                                                                                                                                                                              |
| 9. Implementation Readiness Validation       | Conditional pass — release approved pending the heading-structure correction, verified by the quality lead without a full re-review.                                                                                                                                                                                         |
| 10. Continuous Quality Improvement           | Recommend adding an automated heading-structure check to the production checklist so this finding does not recur.                                                                                                                                                                                                            |

*Table 11.5. Quality Intelligence Specification, all ten domains — AI Practitioner Certification Program. Domains marked “routine” reflect standard practice rather than a case-specific finding.*

As in Chapter 10, a conditional pass with one specific corrective action (Domain 8) is more useful than a blanket approval — and it's worth noting plainly that both chapters' quality reviews surface an accessibility defect rather than an instructional-content one. That is not a coincidence manufactured for this book: heading-structure and navigation issues are among the most commonly reported defect categories in real learning-platform content (WebAIM, 2024), which is exactly why accessibility is a distinct checklist item (Section 2.6) rather than an assumption folded into general review.

## 11.6 Stage 5: Knowledge Intelligence Engine

The governance exception logged in Section 11.4 becomes a reusable organizational asset, not just a resolved incident. Table 11.6 covers all ten knowledge domains from Section 2.7.

| Domain (Section 2.7)                  | AI Practitioner Certification Program                                                                                                                                                                                                                                                                                            |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Knowledge Acquisition (routine)    | Standard capture of the governance exception and accessibility finding at the close of the production cycle.                                                                                                                                                                                                                     |
| 2. Knowledge Organization (routine)   | Filed under “AI Instructional Production, compliance content risk” and “Accessibility, heading structure” in the center for teaching and learning’s repository.                                                                                                                                                                  |
| 3. Knowledge Validation (routine)     | Reviewed by the quality lead before publication to the repository, confirming a reproducible pattern rather than a one-off.                                                                                                                                                                                                      |
| 4. Knowledge Asset Management         | The revised AI-assisted-feedback scenario, with its explicit approved-tool qualification, retained as a reusable template for future compliance-adjacent scenario drafting.                                                                                                                                                      |
| 5. Pattern and Trend Analysis         | AI-drafted content describing a generally helpful AI use case recurrently omits the compliance qualification that would make the advice safe to follow as written — the same underlying pattern Chapter 10 identifies with regulatory figures, here applied to a federal data-privacy requirement instead of a dollar threshold. |
| 6. Best Practices and Lessons Learned | Any AI-drafted content endorsing an AI use case involving student, patient, or client data must include the applicable data-handling qualification as literal source text from the authorizing office, never generated or paraphrased by AI, even when the general advice reads as reasonable.                                   |
| 7. Decision Intelligence              | This finding informs the decision to require literal source text for any legally mandated data-handling content across every future university training initiative, not only AI practitioner certification.                                                                                                                      |
| 8. Innovation and Improvement         | Proposed pilot: an automated pre-review check flagging AI-drafted content that recommends an AI tool or workflow without a literal citation to the current approved-tool list.                                                                                                                                                   |

|                                                        |                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9. Knowledge Sharing and Reuse                         | Rule applied to the AI Governance Specification for the following year's certification cohort before drafting begins, rather than rediscovered during review.                                                                                                                                         |
| 10. Organizational Learning and Capability Development | Across production cycles, the center for teaching and learning's AI-assisted drafting can shift from catching compliance omissions during review to preventing them through an upstream content boundary — the same organizational-learning trajectory Chapter 10 demonstrates in a different sector. |

*Table 11.6. Knowledge Intelligence Specification, all ten domains — AI Practitioner Certification Program. Domains marked “routine” reflect standard practice rather than a case-specific entry.*

This is the organizational-learning claim from Section 2.7 made concrete a third time, in a third register: the lesson does not live only in one reviewer's memory. It becomes a standing governance rule the next cohort inherits automatically.

## 11.7 Before/After: Governed vs. Ad Hoc AI Use

Table 11.7 contrasts this governed certification redesign with how the same project frequently unfolds without an operating system like IDOS – not because ad hoc AI use is reckless by intent, but because without an explicit boundary, decisions like the data-handling question in Section 11.4 have no guaranteed checkpoint at which they are caught.

| Dimension           | Ad hoc AI-assisted development                                          | Under IDOS                                                                     |
|---------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Compliance accuracy | Depends on whichever reviewer happens to notice a missing qualification | Explicit governance rule; data governance review required by policy, not habit |
| Accountability      | Unclear who approved the final scenario language                        | Traceable decision log from design through release                             |
| Reusable learning   | The lesson lives with whoever caught the omission                       | Logged as a governance rule inherited by future cohorts                        |

|               |                                         |                                                     |
|---------------|-----------------------------------------|-----------------------------------------------------|
| Quality check | Often informal, by whoever is available | Independent review, separate from the drafting team |
|---------------|-----------------------------------------|-----------------------------------------------------|

*Table 11.7. Illustrative comparison of governed versus ad hoc AI-assisted development for this case.*

This comparison is illustrative, not a controlled study – no claim is made about how often ad hoc processes actually fail this way in practice, only that IDOS's specific design closes a gap that has no guaranteed safeguard without it.

## 11.8 What This Case Does, and Does Not, Show

This chapter demonstrates that IDOS's artifacts and checkpoints generalize beyond the sector and risk type Chapter 10 used to introduce them: a content boundary can be built around a federal data-privacy requirement as readily as a regulatory dollar figure, a governance exception can be logged and resolved by a subject-matter authority outside the instructional design team (here, a data governance office rather than a compliance officer), and a lesson caught once becomes a rule the next cohort inherits, regardless of sector. It also demonstrates something specific to this case: the failure mode that most needs a content boundary is not always a wrong claim, it can be an omission inside otherwise reasonable, encouraging advice, which is harder for a reviewer to catch by instinct than an outright error.

It does not demonstrate that universities using IDOS produce measurably better AI practitioner certification, catch more compliance omissions, or reduce data-privacy risk more than they would under a well-run ad hoc process with a diligent reviewer. No such comparative data exists for this framework. Like Chapter 10, this chapter is a composite built to demonstrate architectural coherence, that the pieces fit together and produce sensible, specific artifacts, not evidence of outcomes. Readers evaluating IDOS for adoption should treat the

Limitations and Future Research section in the back matter as the more important word on what remains unproven.

## 11.9 Chapter Summary

Run end to end, IDOS's five engines produced, for this illustrative case: a Human Design Specification that made a legally consequential content boundary explicit before drafting began, requiring both an accurate approved-tool qualification and sign-off from the data governance office; an AI Governance Specification that elevated risk and compliance to a genuine case-specific decision, reflecting the real stakes of AI-assisted work with student data; a production stage in which that boundary caught an AI draft, produced by the Content Development and Assessment Design Agents (Section 2.9), that quietly omitted a federal data-privacy qualification inside otherwise reasonable advice; an independent quality review that produced a specific accessibility finding rather than a blanket pass; and a knowledge-repository entry converting the catch into a standing rule inherited by future certification cohorts. Together with Chapter 10, this book now demonstrates three of Chapter 8's six design objects, Module, Microlearning, and Competency-Based Program, across two sectors, using the same composite technique in both chapters, held to the same honest standard about what each does and does not prove.

BACK MATTER

**Appendices, References, and Index**

## Appendix A

### Enterprise Implementation Toolkit

*This appendix contains fillable templates for the four enterprise management documents introduced in Part II: the Enterprise Readiness Assessment (Chapter 4), the Enterprise Governance Plan (Chapter 5), the Operational Execution Framework (Chapter 6), and the Enterprise Performance Management Plan (Chapter 7). Each template follows the same pattern: organization information, the core content sections specific to that document, and an approval block. Copy the relevant template into your organization's document system and adapt field labels, review cycles, and approval roles to your context, nothing here should be treated as a fixed form.*

#### A.1 Enterprise Readiness Assessment (ERA)

Completed before implementation begins (Chapter 4). Reassess at the end of each implementation phase.

| Field                | Entry |
|----------------------|-------|
| Organization / unit  |       |
| Assessment date      |       |
| Assessment lead      |       |
| Implementation scope |       |

| Readiness pillar | Maturity level (1–5) | Evidence / notes |
|------------------|----------------------|------------------|
| Leadership       |                      |                  |
| Governance       |                      |                  |
| Workforce        |                      |                  |

|            |  |  |
|------------|--|--|
| Operations |  |  |
| Technology |  |  |

*Table A.1. ERA readiness scorecard (see Table 4.2 for the five-level maturity model).*

| Capability gap | Priority (Critical / Important / Developmental) | Recommended action |
|----------------|-------------------------------------------------|--------------------|
|                |                                                 |                    |
|                |                                                 |                    |
|                |                                                 |                    |

*Table A.2. Capability gap register (see Table 4.3).*

*Implementation recommendation:* ☐ Ready to proceed ☐ Proceed with conditions  
☐ Additional preparation required

*Approved by:* \_\_\_\_\_ *Role:*  
 \_\_\_\_\_ *Date:* \_\_\_\_\_

## A.2 Enterprise Governance Plan (EGP)

Established after the ERA and before operational implementation (Chapter 5). Review at least annually.

| Governance domain | Owner | Primary decision rights |
|-------------------|-------|-------------------------|
| Leadership        |       |                         |
| Operations        |       |                         |
| AI                |       |                         |
| Quality           |       |                         |
| Knowledge         |       |                         |

*Table A.3. Governance responsibility matrix (see Table 5.1).*

| AI policy domain                    | Policy owner | Last reviewed |
|-------------------------------------|--------------|---------------|
| Acceptable use                      |              |               |
| Human oversight                     |              |               |
| Transparency and disclosure         |              |               |
| Data privacy and protection         |              |               |
| Intellectual property and copyright |              |               |
| Accessibility and inclusion         |              |               |
| Security                            |              |               |
| Quality assurance                   |              |               |
| Risk and incident management        |              |               |
| Continuous improvement              |              |               |

Table A.4. AI policy tracking log (see Table 5.4).

*Approved by:* \_\_\_\_\_ *Role:*  
 \_\_\_\_\_ *Date:* \_\_\_\_\_

### A.3 Operational Execution Framework (OEF)

Documents how a specific instructional workflow applies the six-stage model (Chapter 6). Complete one per major initiative or program.

| Workflow stage | Owner | Governance checkpoint | Target date |
|----------------|-------|-----------------------|-------------|
| Plan           |       |                       |             |
| Design         |       |                       |             |
| Develop        |       |                       |             |
| Validate       |       |                       |             |
| Implement      |       |                       |             |

|       |  |  |  |
|-------|--|--|--|
| Learn |  |  |  |
|-------|--|--|--|

Table A.5. Workflow tracking template (see Table 6.1).

*Design object (course / module / lesson / pathway / competency program / microlearning — see Table 8.1):* \_\_\_\_\_

*Approved by:* \_\_\_\_\_ *Role:*  
\_\_\_\_\_ *Date:* \_\_\_\_\_

**A.4 Enterprise Performance Management Plan (EPMP)**

Establishes ongoing measurement across the six performance domains (Chapter 7). Review dashboards monthly; conduct a comprehensive review at least annually.

| Performance domain     | Key indicator | Evidence source | Review frequency |
|------------------------|---------------|-----------------|------------------|
| Strategic              |               |                 |                  |
| Governance             |               |                 |                  |
| Operational            |               |                 |                  |
| Human-AI collaboration |               |                 |                  |
| Quality                |               |                 |                  |
| Knowledge              |               |                 |                  |

Table A.6. Enterprise performance indicator log (see Table 7.1 and Table 7.2).

*Approved by:* \_\_\_\_\_ *Role:*  
\_\_\_\_\_ *Date:* \_\_\_\_\_

## Appendix B

### Quick Implementation Guide

*This appendix summarizes the implementation sequence from Part II in a single reference. It is a navigation aid, not a replacement for Chapters 4–7 – use it to orient a new team member or a first planning conversation, then work from the full chapters and the Appendix A templates for the actual implementation.*

#### B.1 The Seven-Step Sequence

| Step       | Primary question               | Key deliverable                        | Book reference                 |
|------------|--------------------------------|----------------------------------------|--------------------------------|
| 1. Assess  | Is the organization ready?     | Enterprise Readiness Assessment        | Ch. 4; Appendix A.1            |
| 2. Govern  | Who decides what?              | Enterprise Governance Plan             | Ch. 5; Appendix A.2            |
| 3. Design  | What are we building, and how? | Design object blueprint                | Ch. 2, 8; Appendix A.3         |
| 4. Operate | How does daily work run?       | Operational Execution Framework        | Ch. 6; Appendix A.3            |
| 5. Measure | Is it working?                 | Enterprise Performance Management Plan | Ch. 7; Appendix A.4            |
| 6. Learn   | What did we discover?          | Knowledge repository entry             | Ch. 2 (KIE); Ch. 10 case study |
| 7. Improve | What changes next cycle?       | Updated governance / workflow          | Ch. 5, 6, 9                    |

*Table B.1. The seven-step implementation sequence and where each step is covered in full.*

This is a cycle, not a line: Step 7 feeds back into Step 1 for the organization's next initiative, consistent with the Knowledge Intelligence Engine's role in Chapter 2 and the continuous-improvement principle running through Chapters 4, 5, 7, and 9.

#### B.2 Before You Begin

Confirm these five conditions before starting Step 1; their absence is the most common reason implementations stall.

- Executive sponsorship is named, not assumed.
- At least a draft AI governance policy exists, even if incomplete (Table 5.4).
- Someone has explicit ownership of each of the five operational engines for the pilot initiative (Table 6.2).
- A single design object has been chosen for the pilot – not “all instructional content,” one course or one module (Table 8.1).
- There is a plan to capture what is learned, not just to complete the pilot (Chapter 2, Knowledge Intelligence Engine).

### B.3 Implementation Checklist

| Activity                                               | Done                     |
|--------------------------------------------------------|--------------------------|
| Enterprise Readiness Assessment completed              | <input type="checkbox"/> |
| Executive sponsor identified                           | <input type="checkbox"/> |
| Governance structure and AI policy domains established | <input type="checkbox"/> |
| Pilot design object selected and scoped                | <input type="checkbox"/> |
| Operational workflow documented (six stages)           | <input type="checkbox"/> |
| Independent quality review process confirmed           | <input type="checkbox"/> |
| Performance indicators selected                        | <input type="checkbox"/> |
| Knowledge-capture process established                  | <input type="checkbox"/> |

*Table B.2. Pre-implementation checklist.*

## Appendix C

### Glossary of Key Terms

*Terms are defined as used within IDOS and may differ from usage in other frameworks. Entries are alphabetized; the chapter in parentheses is where the term is introduced in full.*

#### How These Terms Relate

Several terms in this book, framework, operating system, architecture, engine, operational object, governance, and workflow, are related but not interchangeable, and two of them deliberately carry both a general meaning and a specific, capitalized one. This table shows the relationship before the alphabetical entries define each term individually.

| Term                         | What it refers to, and how it relates to the others                                                                                                                                                                                                                                                                                                       |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Framework / operating system | Both describe IDOS itself, the whole thing this book is about. “Framework” names what kind of thing IDOS is (a methodology organizations adopt); “operating system” names how it works (a foundational layer other instructional work runs on top of, per the Introduction’s note on the name). Same referent, two descriptions, not two competing terms. |
| Architecture                 | IDOS’s internal structure. Used most precisely for the four-layer enterprise architecture (Section 1.5), and reused more loosely elsewhere (“governance architecture,” “technology architecture”) to mean “how a given part is organized.” When precision matters, this book says which architecture.                                                     |
| Operational engine           | One of the five active components inside the architecture’s second layer, Human Design, AI Governance, AI Instructional Production, Quality Intelligence, Knowledge Intelligence, that do the actual work (Ch. 2).                                                                                                                                        |
| Agent / Core Agent Role      | One of twenty named functional AI roles, four per operational engine, that execute the AI-assisted work within that engine’s authority boundary. Not a specific AI platform, a                                                                                                                                                                            |

|                    |                                                                                                                                                                                                                                                                                                           |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | single platform may perform one or more roles (Ch. 2, section 2.9).                                                                                                                                                                                                                                       |
| Operational object | The single authoritative document one engine produces, the HDS, AGS, AIPS, QIS, or KIS, not the engine itself (Section 3.1). An engine acts; an operational object is what it produces.                                                                                                                   |
| Prompt             | The executable instruction an agent actually runs, built from one or more operational objects. Domains define what decisions must be made; agents define who performs the resulting work; prompts define how an approved decision reaches that agent as something it can execute (Ch. 2, section 2.9.1).  |
| Governance         | Used two ways. As part of an engine's name, “AI Governance Engine,” it means one specific component (Chapter 2). Without “engine” attached, it means the broader property distributed across all five engines, not confined to the AGE alone (Section 3.3). Most uses in this book are the broader sense. |
| Workflow           | Used two ways. Capitalized or paired with “six-stage,” it means the specific Plan–Design–Develop–Validate–Implement–Learn sequence (Chapter 6). Lowercase and unpaired, it means any sequence of operational steps generally (a review workflow, a production workflow).                                  |

*Table C.1. How the book's core structural terms relate to one another.*

## A

| Term                              | Definition                                                                                                                                                                                                                        |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Agent (Core Agent Role)           | One of twenty functional AI roles distributed across the five operational engines, four per engine, executing work within that engine's authority boundary. Not a specific AI platform (Ch. 2, section 2.9).                      |
| AI Governance Engine (AGE)        | The operational engine that translates approved instructional intent into enforceable rules for AI participation (Ch. 2).                                                                                                         |
| AI Governance Specification (AGS) | The authoritative document produced by the AI Governance Engine (Ch. 2).                                                                                                                                                          |
| AI Policy Framework               | The ten policy domains, acceptable use, human oversight, transparency, data privacy, IP, accessibility, security, quality, risk management, continuous improvement, that together constitute an enterprise AI policy set (Ch. 5). |

|                                                  |                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI Instructional Production Engine (AIPE)        | The operational engine that executes governed AI-assisted production (Ch. 2).                                                                                                                                                                                                                                                                             |
| AI Instructional Production Specification (AIPS) | The authoritative document produced by the AI Instructional Production Engine (Ch. 2).                                                                                                                                                                                                                                                                    |
| AI Tool Evaluation Criteria                      | The eight criteria, capability fit, bias and fairness, data handling and privacy, security, transparency and explainability, accessibility of outputs, cost and sustainability, vendor accountability, used to assess a candidate AI tool for a specific agent role before adoption, distinct from governing an already-adopted tool (Ch. 4, Table 4.1a). |

## C

| Term                           | Definition                                                                                                                   |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Capability gap                 | The difference between an organization's current readiness and the readiness required for successful implementation (Ch. 4). |
| Competency Mastery Framework   | The eight-stage model (Identify–Advance) governing competency-based program design (Ch. 8).                                  |
| Content boundary               | A governance rule restricting what AI may generate, add, or infer within instructional content (Ch. 10, 11).                 |
| Cross-Engine AI Agent Taxonomy | The full twenty-role table organizing Core Agent Roles by operational engine (Ch. 2, section 2.9).                           |

## D

| Term                      | Definition                                                                                                                                      |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Decision authority matrix | The table assigning primary ownership of each recurring decision category to one governance role (Ch. 5, Table 5.2).                            |
| Quality checkpoint        | A formal checkpoint requiring approval before work proceeds to the next workflow stage (Ch. 6).                                                 |
| Design object             | One of the six things IDOS can be used to design: course, module, lesson, learning pathway, competency-based program, or microlearning (Ch. 8). |

## E

| Term | Definition |
|------|------------|
|------|------------|

|                                               |                                                                                                                                                                                                                            |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Enterprise Governance Plan (EGP)              | The organization-level governance document covering roles, decision rights, and policy (Ch. 5; Appendix A.2).                                                                                                              |
| Enterprise Performance Management Plan (EPMP) | The document establishing ongoing measurement across the six performance domains (Ch. 7; Appendix A.4).                                                                                                                    |
| Enterprise Prompt Architecture (EPA)          | The mechanism that translates approved operational specifications into executable prompts for AI agents. Specifications define enterprise decisions; the EPA operationalizes them for AI execution (Ch. 2, section 2.9.1). |
| Enterprise Readiness Assessment (ERA)         | The pre-implementation diagnostic across five readiness pillars (Ch. 4; Appendix A.1).                                                                                                                                     |

## H

| Term                             | Definition                                                                                                                                   |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Human Design Engine (HDE)        | The operational engine that establishes instructional intent before AI-assisted production begins; the instructional kernel of IDOS (Ch. 2). |
| Human Design Specification (HDS) | The authoritative document produced by the Human Design Engine (Ch. 2).                                                                      |
| Human-in-Governance principle    | The design commitment that AI may inform decisions anywhere in the system but does not govern the system (Ch. 3).                            |

## I

| Term                                              | Definition                                                                                                                                         |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| IDOS                                              | The Human–AI Instructional Design Operating System — the enterprise framework presented in this book (Ch. 1).                                      |
| Instructional kernel                              | The architectural role of the Human Design Engine: the authoritative source of instructional reasoning that every other engine depends on (Ch. 2). |
| Instructional Modernization Maturity Model (IMMM) | The four-level scale (Refresh, Enhance, Redesign, Transform) used to scope a modernization project (Ch. 9).                                        |

## K

| Term                                       | Definition                                                                                                                |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Knowledge Intelligence Engine (KIE)        | The operational engine that captures validated experience and converts it into reusable organizational knowledge (Ch. 2). |
| Knowledge Intelligence Specification (KIS) | The authoritative document produced by the Knowledge Intelligence Engine (Ch. 2).                                         |

## L

| Term                        | Definition                                                                                                                                  |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Learner Journey             | The eight-stage model (Enter–Advance) governing learning pathway design (Ch. 8).                                                            |
| Legacy Asset Classification | The five-way decision (Preserve, Enhance, Redesign, Retire, Replace) applied to existing instructional assets during modernization (Ch. 9). |
| Lesson Flow                 | The eight-stage model (Connect–Extend) governing single-lesson design (Ch. 8).                                                              |

## O

| Term                                  | Definition                                                                                                                                                                |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operational engine                    | One of the five core organizational capabilities of IDOS: Human Design, AI Governance, AI Instructional Production, Quality Intelligence, Knowledge Intelligence (Ch. 2). |
| Operational Execution Framework (OEF) | The document coordinating daily instructional workflow (Ch. 6; Appendix A.3).                                                                                             |

## P

| Term                                      | Definition                                                                                                                                                                                                      |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Performance Support Framework             | The eight-stage model (Identify–Improve) governing microlearning and performance-support design (Ch. 8).                                                                                                        |
| Principles of Instructional Modernization | The ten enterprise principles that guide instructional modernization decisions, ensuring modernization efforts are evidence-based, proportional, learner-centered, governed, and continuously improved (Ch. 9). |
| Prompt                                    | A structured operational instruction, built from one or more approved specifications, that communicates an enterprise                                                                                           |

|                              |                                                                                                                                                                                                   |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | decision to an AI agent for a defined task — not a casual conversational request (Ch. 2, section 2.9.1).                                                                                          |
| Prompt inheritance principle | The design commitment that a prompt draws context only from the specification, or specifications, its assigned task actually requires, not from all five by default (Ch. 2, section 2.9.1).       |
| Prompt template              | The reusable enterprise structure from which an individual prompt is instantiated for a specific task; governed and validated like any other Knowledge Intelligence asset (Ch. 2, section 2.9.1). |

## Q

| Term                                     | Definition                                                                                                |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Quality Intelligence Engine (QIE)        | The operational engine that independently validates instructional products before implementation (Ch. 2). |
| Quality Intelligence Specification (QIS) | The authoritative document produced by the Quality Intelligence Engine (Ch. 2).                           |

## R

| Term             | Definition                                                                                                                        |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Readiness pillar | One of five domains, leadership, governance, workforce, operations, technology, evaluated during implementation planning (Ch. 4). |

## S

| Term                   | Definition                                                                                                               |
|------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Single source of truth | The principle that each operational engine's specification is the one authoritative record for its domain (Ch. 3).       |
| Specification          | The authoritative document any operational engine produces (HDS, AGS, AIPS, QIS, KIS) — see each engine's entry (Ch. 2). |

## T

| Term         | Definition                                                                                                                                           |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Traceability | The ability to follow an instructional decision from its origin in one operational engine through every downstream engine it affects (Ch. 2, Ch. 3). |

V

| Term       | Definition                                                                                                                                                              |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Validation | Independent confirmation of implementation readiness, performed by a reviewer not involved in producing the material under review (Ch. 2, Quality Intelligence Engine). |

W

| Term     | Definition                                                                                                                                |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Workflow | The six-stage sequence, Plan, Design, Develop, Validate, Implement, Learn, that governs daily instructional execution under IDOS (Ch. 6). |

Limitations and Future Research

This section states plainly what this book has not demonstrated, alongside what it has. Readers evaluating IDOS for adoption should weigh this section as seriously as any other part of the book — arguably more seriously, since the earlier chapters make the case for the framework and this one is where that case is checked.

What This Book Has Not Shown

No implementation of IDOS has been evaluated in published, peer-reviewed research at the time of writing. The framework's five operational engines, its agent taxonomy, its governance model, and its enterprise documents are design proposals grounded in established instructional-design, learning-science, and AI-governance literature (cited throughout Parts I–IV), not empirical findings from organizations that have used them. The worked cases in Chapters 10 and 11 are both composites, synthesized from well-documented, common patterns in AI-assisted compliance training and higher-education certification respectively; neither is a report of a real

deployment, and no outcome data, completion rates, error reduction, cost savings, compliance improvement, are claimed or should be inferred from either. Together they show the framework's artifacts are producible, comprehensively and specifically, across two sectors and three of Chapter 8's six design objects, using the same composite technique in both — evidence of architectural coherence, not evidence that IDOS causes better outcomes.

Several specific claims in this book are architectural reasoning rather than demonstrated results, and are flagged again here for clarity:

- That IDOS improves instructional quality, consistency, or speed relative to well-run ad hoc AI-assisted development has not been tested comparatively. Chapter 10's before/after table (Table 10.1) is illustrative of what the framework is designed to prevent, not evidence of how often such failures occur in practice without it.
- That the governance overhead IDOS introduces, quality checkpoints, independent quality review, mandatory knowledge capture, is worth its cost in time and coordination has not been quantified. For a small instructional team, this overhead may be disproportionate to the risk it manages; the book does not establish where that threshold lies.
- That the five-engine architecture, and the twenty-role agent taxonomy built on it, generalizes beyond the two sectors and three design objects demonstrated in Chapters 10 and 11, corporate compliance training and higher-education certification, both as fictional composites, to contexts such as small nonprofits, individual consultants, or organizations without an external compliance authority (a data governance office, a regulatory body) to anchor governance around, is untested.

- That the specific compliance details in Chapters 10 and 11, the Bank Secrecy Act threshold, FERPA data-handling requirements, remain current is not something this book can guarantee at the time a reader encounters it; both chapters flag this directly, but the underlying point is general: any organization adapting these examples must verify current requirements against its own authoritative source, not against this book.
- That the specific maturity levels, readiness pillars, and scoring approaches in Chapters 4 and 9 produce consistent, reliable assessments across different raters has not been validated; they are proposed as a starting structure, not a psychometrically tested instrument.

## Scope Limitations

The book focuses on instructional design and enterprise learning contexts. Its governance model draws on general AI-governance guidance (NIST, 2023; OECD, 2019; ISO, 2023) but does not address sector-specific regulatory regimes, healthcare, financial services, government, in the depth those contexts would require for direct implementation. Organizations in heavily regulated industries should treat the governance chapters as a starting architecture to be reviewed against applicable law, not as sufficient compliance guidance on their own.

The book's citations lean on well-established literature in instructional design (Dick et al., 2014; Gagné et al., 2005), learning science (Mayer, 2021; Sweller et al., 1998), and quality management (Juran & Godfrey, 1999) rather than on a systematic literature review of the rapidly evolving AI-in-education research base, much of which postdates the framework's initial design and could not be comprehensively incorporated.

## Future Research Directions

Several lines of inquiry would meaningfully strengthen or revise the claims in this book:

- Longitudinal case studies of organizations implementing IDOS, tracking governance overhead, instructional quality, and organizational learning against a comparable baseline.
- Validation or revision of the readiness and maturity models in Chapters 4 and 9 through structured assessment across multiple organizations and raters.
- Comparative studies isolating the specific contribution of independent quality validation (the Quality Intelligence Engine) versus integrated review, to test whether the separation this book argues for produces measurably better outcomes or primarily better documentation.
- Sector-specific extensions of the governance model for regulated industries, developed in collaboration with domain compliance experts rather than adapted secondhand from general AI-governance frameworks.
- Direct comparison of organizations using IDOS against organizations using ad hoc AI-assisted instructional development, on cost, speed, quality, and error rates — the comparison Chapter 10 gestures toward illustratively but does not conduct empirically.
- Empirical study of whether the twenty-role agent taxonomy in Section 2.9 reflects how organizations actually configure AI tools in practice, or whether real deployments cluster around a smaller functional set.

## A Closing Note

IDOS is offered in this book as a coherent, internally consistent architecture built on established scholarship, not as a proven method.

The distinction matters. Organizations adopting it should do so as an informed bet on a well-reasoned design, pilot it at small scale before committing broadly (consistent with the phased approach in Chapter 4), and treat their own implementation experience as data that should, in the spirit of the Knowledge Intelligence Engine this book describes, be shared back to the field rather than kept private.

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## Index

*Page numbers refer to the arabic pagination of the body of this manuscript (Chapter 1 onward), not the roman-numbered front matter. Entries with more than eight page references are marked “et seq.” rather than listing every occurrence.*

### A

accessibility, 2, 5, 18, 21, 25, 26, 27, 35, et seq.  
 ADDIE, 10, 48, 65, 67, 151, 152  
 Agent, 13, 15, 24, 25, 27, 32, 33, 34, et seq.  
 Agile instructional design, 10, 48  
 AI Governance Engine, 2, 6, 14, 20, 36, 38, 40, 47, et seq.  
 AI Governance Specification, 20, 26, 102, 119  
 AI Instructional Production Engine, 22, 34, 38, 47, 66, 84, 93, 113  
 AI Instructional Production Specification, 22, 39, 103, 120  
 AI Policy Framework, 62, 63  
 AI Practitioner Certification, 112, 114, 115, 116, 117, 119, 120, 122, et seq.  
 approved-tool list, 116, 119, 121, 125  
 Assessment Design Agent, 25, 79, 119  
 Asset Refinement, 23, 102, 103, 120

### B

Bank Secrecy Act, 104, 147

### C

capability gap, 54, 131, 140  
 case study, 34, 90, 91  
 Competency Mastery Framework, 83, 85, 87

competency-based, 78, 79, 83, 86, 114, 115, 128, 140, et seq.

compliance training, 97, 147

Content Development Agent, 25, 33, 39, 79, 102, 103, 119

Core Agent Role, 33, 38

Cross-Engine AI Agent Taxonomy, 32

## **D**

data governance office, 113, 115, 116, 117, 119, 120, 121, 127, et seq.

decision authority, 8, 20, 52, 56, 60

## **E**

Enterprise Governance Plan, 58, 130, 131

Enterprise Prompt Architecture, 38, 39, 41, 44

Enterprise Readiness Assessment, 51, 57, 130, 136

EPA, 38, 141

## **F**

FERPA, 114, 115, 116, 117, 118, 121, 147

field note, 55, 61, 68, 91

## **G**

Governance Authority Architecture, 20

## **H**

Human Design Engine, 2, 6, 13, 14, 18, 33, 37, 40, et seq.

Human Design Specification, 15, 17, 18, 23, 39, 52, 98, 100, et seq.

Human-in-Governance, 47

## **I**

Instructional Content Production, 22

instructional kernel, 141

**J**

job aid, 77, 108, 109

**K**

Knowledge Intelligence Engine, 15, 17, 28, 36, 38, 41, 48, 52, et seq.

Knowledge Intelligence Specification, 28, 108, 126

**L**

Learner Journey, 19, 78, 82, 85, 86

Legacy Asset Classification, 90, 91, 95

Lesson Flow, 80, 81, 85, 86, 98, 142

Limitations and Future Research, 146

**M**

microlearning, 34, 78, 79, 84, 85, 86, 108, 109, et seq.

modernization, 42, 63, 88, 89, 90, 91, 92, 94, et seq.

Modernization Maturity Model, 89, 90

Multimedia Production Agent, 25

**N**

NIST, 8, 17, 20, 53, 104, 148

**O**

OECD, 20, 148, 153

Operational Execution Framework, 130, 133

organizational learning, 2, 4, 6, 7, 9, 17, 30, 48, et seq.

OSCQR, 26, 153

**P**

Performance Need Architecture, 18

Performance Support Framework, 85, 109

Production Agent, 25

Prompt inheritance, 39, 40

Prompt template, 40, 68

## **Q**

Quality Intelligence Engine, 14, 17, 23, 25, 26, 35, 38, 60, et seq.

Quality Intelligence Specification, 26, 106, 123

Quality Matters, 26, 153

## **S**

single source of truth, 45

six-stage, 65, 66, 77, 86, 92, 98, 109, 115, et seq.

Success Architecture, 19, 31

## **T**

terminology, 19, 39, 40, 95, 99

## **W**

WCAG, 154